# Biology

# Class 12 (English Medium)

# Solved Exercises — Sindh Textbook Board

13 chapters — 289 questions

# Chapter 15: Homeostasis

## 1. Encircle the correct choice

1. A self-regulating process by which biological systems maintain stability while adjusting to the changing conditions called

✓ (a) Homeostasis

(b) Osmoregulation

(c) Excretion

(d) Biological rhythms

Answer: (a) Homeostasis - Homeostasis is the overall process of keeping the internal environment stable. Osmoregulation and excretion are only parts of it that help in maintaining it.

2. Which does not maintain a stable, homeostatic condition rather it intensifies the change that is happening to the body.

(a) Negative feedback

✓ (b) Positive feedback

(c) Feed back system

(d) Excretion

Answer: (b) Positive feedback - Positive feedback amplifies the original change instead of reversing it. Negative feedback, by contrast, counteracts the change and restores the stable condition.

3. Fresh water organisms also have specialized cells located in their gills and in skin which actively extract Na+, Cl- and Ca+ from external medium and excrete (H+) or basic (HCO3-) for acid base balance in the body fluids are called

(a) Granulocyte

(b) Lymphocyte

✓ (c) Ionocytes

(d) podocyte

Answer: (c) Ionocytes - Ionocytes are the specialized ion-transporting cells of gills and skin in freshwater animals. They take up needed ions from the dilute surroundings and remove H+ or HCO3- to keep acid-base balance.

4. Renal cortex produces an important hormone necessary for the synthesis of RBC's

✓ (a) Erythropoietin

(b) Leukopoietin

(c) Thrombopoietin

(d) Renin

Answer: (a) Erythropoietin - Erythropoietin is released by the kidney, mainly from the renal cortex, when oxygen in blood falls. It stimulates the bone marrow to produce more red blood cells.

5. Glomerulus have specialized cells that are wrapped around blood capillaries that play an active role in preventing plasma proteins from entering the urinary ultrafiltrate called

(a) Epithelial cells

✓ (b) Podocyte cells

(c) Endothelial cells

(d) None of them

Answer: (b) Podocyte cells - Podocytes wrap the glomerular capillaries with their foot processes and form part of the filtration barrier. This barrier allows water and small solutes to pass but holds back plasma proteins and

blood cells.

6. Tubules secretes ions such as hydrogen, potassium, and NH3 into the filtrate while reabsorbing the HCO3- from the filtrate are called

✓ (a) Distal convoluted tubule

(b) Proximal convoluted tubule

(c) Collecting duct

(d) Loop of Henle

Answer: (a) Distal convoluted tubule - The distal convoluted tubule secretes H+, K+ and ammonia into the filtrate and reabsorbs bicarbonate from it. In this way it plays an important role in acid-base and electrolyte balance.

7. Glomerular filtration rate (GFR) is in which proportional to the hydrostatic pressure exerted in glomerulus wall

(a) Indirectly proportional

✓ (b) Directly proportional

(c) Same proportional

(d) High proportional

Answer: (b) Directly proportional - When hydrostatic pressure in the glomerulus rises, more fluid is forced through the filtration membrane, so GFR increases. When the pressure falls, GFR also falls.

8. Another compound that increases the osmotic gradient in inner medulla is the

✓ (a) Urea

(b) Water

(c) Sulphates

(d) Phosphates

Answer: (a) Urea - Along with NaCl, urea accumulates in the inner medulla and increases its osmotic gradient. This gradient helps the collecting duct to reabsorb water and produce concentrated urine.

9. The inflammation in kidneys due to irritation of kidney stones is called

✓ (a) Lithonephritis

(b) Edema

(c) Sarcoma

(d) Encephalitis

Answer: (a) Lithonephritis - Lithonephritis is the inflammation of the kidney caused by the irritation of kidney stones. Edema is swelling due to fluid, sarcoma is a cancer, and encephalitis is inflammation of the brain.

10. If the surrounding environment becomes hot or a person do strenuous physical activity the heat produced inside the body and raises body temperature above suitable limit this condition is called

(a) Exothermia

✓ (b) Hyperthermia

(c) Hypothermia

(d) Endothermia

Answer: (b) Hyperthermia - Hyperthermia is the rise of body temperature above the normal suitable limit due to external heat or heavy physical activity. Hypothermia is the opposite condition, in which body temperature falls below normal.

## 2. Write short answers of the following questions:

Question: 1. Why the physiological integration of internal body

environment is important for living organisms?

Answer: Cells can work properly only when temperature, pH, water, salts and nutrients around them stay within narrow limits. Physiological integration coordinates organs such as the kidney, liver, lungs and endocrine glands so that this internal environment remains stable. Without it, enzymes would fail and the organism could not survive changes in the external environment.

Question: 2. Why it is important that positive and negative feedback mechanisms counter act upon each other?

Answer: Negative feedback keeps most body variables stable by reversing any change, while positive feedback rapidly completes certain processes such as blood clotting and childbirth. The two must balance and counteract each other so that a useful positive response does not continue endlessly and upset homeostasis. This balance allows both stability and timely, controlled change in the body.

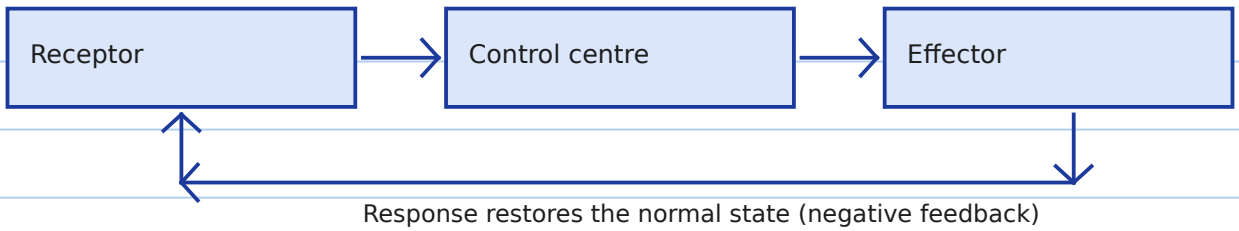



Fig.: Feedback (labelled diagram)

Question: 3. How the aquatic Osmoregulators overcome the osmoregulatory problems?

Answer: Freshwater animals face continuous entry of water and loss of salts, so they excrete large amounts of dilute urine and actively absorb Na+, Cl- and other ions through ionocytes in the gills and skin. Marine animals face the opposite problem of water loss and salt gain, so they drink seawater, excrete excess salt through gills and produce very little concentrated urine. In both cases excretion and active ion transport keep the internal osmotic concentration constant.

Question: 4. Why animals excrete different types of excretory waste?

Answer: The type of nitrogenous waste depends on the availability of water and the habitat of the animal. Ammonia is highly toxic but needs much water for its removal, so it is excreted mainly by aquatic animals, while urea is less toxic and needs moderate water, so it is excreted by mammals. Uric acid is almost non-toxic and needs very little water, so it suits birds, reptiles and desert animals.

Question: 5. What is the use of counter current mechanism in kidney?

Answer: The counter current mechanism in the loop of Henle builds a strong osmotic gradient in the medulla, from the cortex towards the inner medulla. This gradient allows water to be drawn out of the collecting duct under the control of ADH. As a result, the kidney can produce urine more concentrated than blood and conserve water when the body needs it.

Question: 6. What is the role of kidney as an endocrine organ?

Answer: Besides excretion, the kidney secretes important hormones, so it also acts as an endocrine organ. It produces erythropoietin, which stimulates red blood cell formation in bone marrow, and renin, which helps regulate blood pressure and salt balance. It also activates vitamin D, which is needed for calcium absorption.

Question: 7. How kidney stones are formed?

Answer: Urine contains dissolved minerals and salts such as calcium oxalate, calcium phosphate and uric acid. When urine becomes too concentrated, or these substances increase, they crystallize and gradually deposit in the kidney or urinary tract to form stones. Low water intake, excess salts in diet and urinary infections increase the risk of stone

formation.

Question: 8. What is the impact over the human body if kidneys suddenly or gradually loss the filtering abilities?

Answer: When kidneys lose their filtering ability, urea and other wastes accumulate in the blood and water and salt balance is disturbed. This causes nausea, swelling, high blood pressure, anaemia, weakness and disturbance of acid-base balance. In complete kidney failure the condition becomes life-threatening unless dialysis or transplant is provided.

Question: 9. What are the problems associated with kidney transplant?

Answer: The main problem is rejection, because the immune system of the recipient recognizes the donor kidney as foreign and attacks it, so immunosuppressive drugs are needed for life. These drugs increase the risk of infections and other side effects. Other problems include shortage of suitable donors, the need for close tissue matching, high cost and possible failure of the transplanted kidney after some years.

Question: 10. What is hypothermia? How body overcome this condition?

Answer: Hypothermia is the condition in which body temperature falls below the normal limit, usually due to prolonged exposure to cold. The body overcomes it by shivering, which produces heat through rapid muscle contractions, and by constriction of skin blood vessels, which reduces heat loss. Behavioural responses such as seeking shelter and wearing warm clothes also help restore normal temperature.

## 3. Write detailed answers of the following questions:

Question: 1. Explain the detailed structure of nephron with labelled

diagram

Answer: A nephron is the structural and functional unit of the kidney, and each kidney contains about a million nephrons. It begins with the renal corpuscle, made of the glomerulus, a tuft of capillaries, surrounded by the cup-shaped Bowman's capsule whose inner layer is formed of podocytes; here ultrafiltration of blood takes place. The filtrate then passes through the proximal convoluted tubule, where most glucose, amino acids, salts and water are reabsorbed, then through the loop of Henle, whose descending and ascending limbs create the counter current gradient, and then through the distal convoluted tubule, where secretion of H+, K+ and drugs and further reabsorption occur. Finally the nephron opens into the collecting duct, which carries urine to the renal pelvis while reabsorbing water under ADH control.

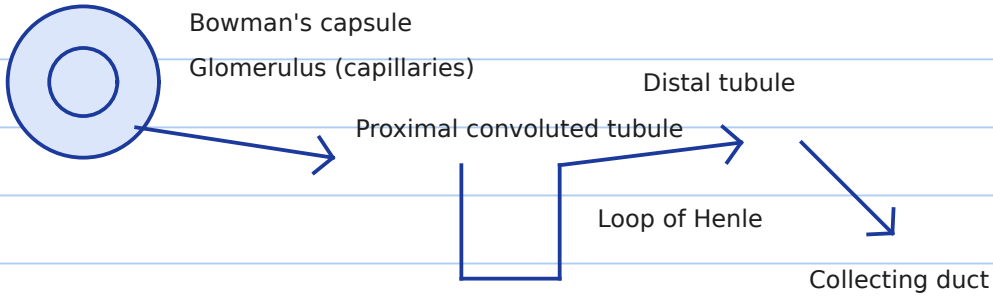



Fig.: Nephron (labelled diagram)

Question: 2. Explain the functions of kidney

Answer: The kidney performs several vital functions for homeostasis. It excretes nitrogenous wastes such as urea, excess salts and drugs in urine, regulates the water and osmotic balance of blood, and maintains acid-base balance by secreting H+ and reabsorbing bicarbonate. It also regulates blood pressure through renin and fluid volume control, secretes erythropoietin for red blood cell formation, and activates vitamin D for calcium metabolism.

Question: 3. Explain the causes of kidney failure and its treatment

Answer: Kidney failure may be sudden or gradual and is caused by

conditions such as uncontrolled diabetes and high blood pressure, glomerular inflammation, kidney stones, urinary tract obstruction, infections and toxic drugs that damage the nephrons. As filtration fails, wastes and fluids accumulate in the body and disturb blood chemistry. Treatment includes controlling the cause and diet, haemodialysis or peritoneal dialysis and continuous renal replacement therapy to artificially filter the blood, and in severe permanent failure, kidney transplant from a suitable donor.

# Chapter 16: Support and Movement

## 1. Encircle the correct choice.

1. The type of bone cell that tears down bone during the building and remodeling process are

(a) Osteocytes

(b) Osteoblasts

✓ (c) Osteoclasts

(d) Bone lining cells

Answer: (c) Osteoclasts - Osteoclasts are large cells that break down and resorb old bone tissue during growth and remodeling. Osteoblasts do the opposite work of building new bone, while osteocytes maintain the mature bone.

2. During bone formation, as the periosteum calcifies, it gives rise to a thin plate of compact bone called

(a) Periosteal bud

✓ (b) Periosteal bone collar

(c) Primary ossification center

(d) Epiphyseal plate

Answer: (b) Periosteal bone collar - During bone formation the periosteum around the shaft calcifies and forms a thin plate of compact bone called the periosteal bone collar. This collar supports the developing bone and is later thickened as ossification proceeds.

3. The ring like shape of the girdle is due to the joining of coxal bone with the sacrum of vertebral column at anterior side and below by a joint in

between pubic part called

✓ (a) Pubic symphysis

(b) Iliac junction

(c) Pelvic symphysis

(d) Syndesmosis

Answer: (a) Pubic symphysis - The two coxal bones join the sacrum behind and meet each other in front at the pubic symphysis, a cartilaginous joint. This arrangement completes the ring-like pelvic girdle that supports the trunk and protects the pelvic organs.

4. Cartilage is a soft flexible form of connective tissue surrounded by a layer called

(a) Myocardium

(b) periosteum

(c) Osteoderm

✓ (d) Perichondrium.

Answer: (d) Perichondrium - Cartilage is covered externally by a dense connective tissue layer called the perichondrium, which nourishes it and helps in its growth and repair. Periosteum is the corresponding covering layer of bone, not of cartilage.

5. An antagonist muscle is relaxed while the other side opposite to antagonist muscle

✓ (a) Contracted

(b) Relaxed

(c) No change

(d) Extremely relaxed

Answer: (a) Contracted - Muscles work in antagonistic pairs, so when one

muscle, the agonist, contracts to produce a movement, the opposite antagonist muscle relaxes. For example, during flexion of the forearm the biceps contracts while the triceps relaxes.

6. The stage when old bone tissues are removed and gets replaced with new ones that joins both fractured sides called

✓ (a) Bone remodeling stage

(b) Bone callus formation

(c) Cartilaginous callus formation

(d) Hematoma formation

Answer: (a) Bone remodeling stage - In the final stage of fracture repair, osteoclasts remove old and dead bone and osteoblasts lay down new bone that firmly joins both fractured sides. The bone is gradually remodeled to its original shape and strength.

7. Which is related to the mild to severe grade of stretching or tearing of a ligament that holds the bones of a joint together?

(a) Spasm

(b) Tetany

✓ (c) Sprains

(d) Cramps

Answer: (c) Sprains - A sprain is the stretching or tearing of a ligament that holds the bones of a joint together, ranging from mild to severe. Spasm, tetany and cramps are abnormal conditions of muscles, not of ligaments.

8. Muscle fiber are gathered in many groups each group of fibers is called a fascicle which is surrounded by a membrane called

(a) Sarcolemma

(b) Perimysium

(c) Endomysium

(d) Epimysium

Answer: (b) Perimysium - Each fascicle, a bundle of muscle fibers, is surrounded by a connective tissue membrane called the perimysium. The endomysium covers individual fibers, the epimysium covers the whole muscle, and the sarcolemma is the cell membrane of a fiber.

9. The heart muscles arranged in a branching network where cells are joined together and making a

(a) Syncytium

(b) Sarcoplasmic reticulum

(c) Endoplasmic reticulum

(d) T-tubules

Answer: (a) Syncytium - Cardiac muscle cells branch and are joined end to end by intercalated discs, so the whole tissue works as a single functional unit called a syncytium. This allows the heart to contract in a coordinated wave.

10. These joints hold the bone by dense connective tissue contain collagenous fibers

(a) Fibrous joints

(b) Syndesmosis

(c) Gomphosis

(d) Synovial joints

Answer: (a) Fibrous joints - In fibrous joints the bones are held together directly by dense connective tissue rich in collagen fibers, allowing little or

no movement. Synovial joints, in contrast, have a fluid-filled cavity and allow free movement.

## 2. Write short answers of the following questions:

**Question: 1. How is the cartilage useful for our skeletal system?**

Answer: Cartilage covers the ends of bones at joints, where it reduces friction and absorbs shocks during movement. It also gives shape and flexible support to structures such as the nose, ears and trachea, and in children it forms the model from which bones develop and lengthen. Being soft and flexible, it protects the joints from wear and damage.

**Question: 2. Give a comparison of major types of joints**

Answer: Fibrous joints are joined by dense connective tissue, allow no or very little movement, and are found between skull bones. Cartilaginous joints are joined by cartilage, allow slight movement, and are found between vertebrae and at the pubic symphysis. Synovial joints have a fluid-filled synovial cavity, allow free movement in one or more directions, and include the knee, shoulder and elbow joints.

**Question: 3. How is the healing of fractured bone initiated pain developing?**

Answer: When a bone fractures, blood vessels in the bone and periosteum are torn, causing pain and bleeding that forms a clot called a hematoma around the broken ends. The pain and swelling signal injury, and the hematoma provides the framework in which repair begins. In the next steps a cartilaginous callus joins the broken ends, which is then replaced by bone and finally remodeled.

Question: 4. How to conduct a first aid treatment for joint injuries and fractures?

Answer: The injured part should be kept still and supported, for example with a splint or sling, so that the broken bone or injured joint does not move and cause further damage. Cold packs can be applied to reduce pain and swelling, the limb may be kept raised, and tight bandaging should be avoided. The patient should then be taken to hospital promptly for X-ray, proper alignment and plastering or other treatment.

Question: 5. Give a comparison of types of muscles.

Answer: Skeletal muscles are attached to bones, are striated and voluntary, and produce body movements such as walking and lifting. Smooth muscles are found in the walls of involuntary organs like the gut and blood vessels, are non-striated and work automatically. Cardiac muscle is found only in the heart, is striated but involuntary, and contracts rhythmically throughout life without fatigue.

Question: 6. Why is calcium essential for cross bridge formation?

Answer: When a muscle is stimulated, calcium ions are released from the sarcoplasmic reticulum into the muscle fiber. Calcium binds with troponin, which moves tropomyosin away from the active sites on actin, so the myosin heads can attach to actin and form cross bridges. Without calcium the active sites remain covered and contraction cannot begin.

Question: 7. How the nervous coordination regulates muscle contraction mechanism?

Answer: A motor neuron carries the impulse to the muscle, and at the neuromuscular junction it releases acetylcholine, which generates an

action potential in the muscle fiber membrane. This action potential spreads through the T-tubules and causes the sarcoplasmic reticulum to release calcium ions, which trigger cross bridge formation and contraction. When the nerve impulse stops, calcium is pumped back and the muscle relaxes, so the nervous system controls both the start and the strength of contraction.

Question: 8. How is tetany different from tetanus?

Answer: Tetany is a condition of sustained, painful muscle spasms caused by low calcium in the blood, often due to parathyroid deficiency, and it affects mainly the hands, feet and face. Tetanus is an infectious disease caused by the bacterium Clostridium tetani entering a wound, whose toxin causes severe, continuous contraction of muscles including lockjaw. Thus tetany is a metabolic disorder, while tetanus is a bacterial infection.

Question: 9. Explain the different types of cartilage.

Answer: Hyaline cartilage is glassy and smooth, covers the ends of bones at joints and is found in the nose, trachea and ribs, where it provides support with flexibility. Elastic cartilage contains many elastic fibers, is highly flexible, and is found in the external ear and epiglottis. Fibrocartilage contains dense collagen fibers, is the strongest type, and is found in intervertebral discs and the knee, where it resists compression and absorbs shocks.

## 3. Write detail answers of the following questions:

Question: 1. Explain the structure of skeletal muscles and ultra structure of skeletal muscle fiber with labelled diagram.

Answer: A skeletal muscle is made of bundles of long muscle fibers held

together by connective tissue layers: the endomysium around each fiber, the perimysium around each fascicle, and the epimysium around the whole muscle. Under the sarcolemma, each fiber contains many rod-like myofibrils made of repeating units called sarcomeres, which are the units of contraction. Each sarcomere has thick myosin filaments in the A band and thin actin filaments in the I band, separated by Z lines, and their sliding over each other shortens the fiber and contracts the muscle.

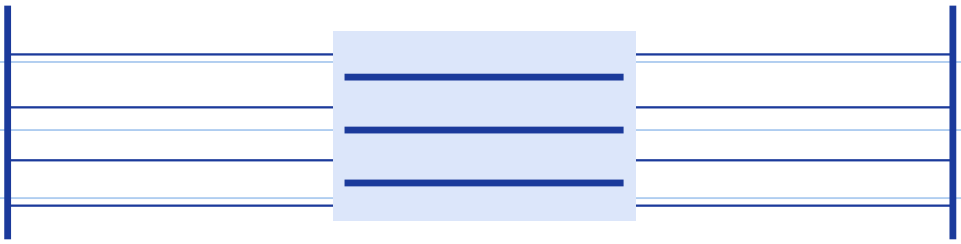



Fig.: Sarcomere (labelled diagram)

Question: 2. Explain the repair process of simple fractures

Answer: Repair of a simple fracture takes place in a fixed sequence of steps. First, bleeding from torn vessels forms a hematoma (blood clot) around the broken ends, which seals the area. Then an internal and external callus of cartilage forms and joins the two sides temporarily. Next, the cartilage of the callus is replaced by trabecular (spongy) bone, making the union firm. Finally, in the remodeling stage, osteoclasts remove dead and excess bone and osteoblasts lay down compact bone, restoring the original shape and strength.

Question: 3. Explain the cross-bridge cycle of muscle contraction with labelled diagram

Answer: In the cross-bridge cycle, calcium first exposes the active sites on actin, and the energized myosin head, carrying ADP and phosphate, attaches to actin to form a cross bridge. The myosin head then bends in a power stroke, pulling the actin filament towards the center of the sarcomere and releasing ADP and phosphate. A new ATP molecule binds to the myosin head, which detaches it from actin, and breakdown of this ATP

re-energizes the head so the cycle can repeat. Repeated cycles slide the actin filaments over myosin and shorten the sarcomere, producing contraction.

# Chapter 17: Nervous Coordination

## 1. Encircle the correct choice.

1. Most of the neurons in the human brain are

(a) Sensory neurons

(b) Motor neurons

✓ (c) Interneurons

(d) Auditory neurons

Answer: (c) Interneurons - Most neurons of the brain are interneurons, which lie entirely within the central nervous system and connect sensory and motor neurons. They are responsible for integration, processing and decision making in the brain and spinal cord.

2. A nervous system can alter activities in its target cells in muscles and gland because

(a) They are electrically coupled by gap junctions.

✓ (b) The target cells have receptor proteins for the signals released by the nervous system.

(c) The nervous system releases signals into the blood to control the target cells.

(d) The target cells that become disconnected from the nervous system rapidly die.

Answer: (b) The target cells have receptor proteins for the signals released by the nervous system. - Neurons release neurotransmitters at their endings, and only those muscle or gland cells that carry matching receptor proteins respond to them. This receptor specificity allows the nervous system to alter the activity of selected target cells.

3. For a neuron with an initial membrane potential at -70mv, an increase in the movement of potassium ions out of that neuron's cytoplasm would result in

(a) Depolarization of the neuron

✓ (b) Hyperpolarization of neuron

(c) The replacement of potassium ions with sodium ions.

(d) The replacement of potassium ions with calcium ions.

Answer: (b) Hyperpolarization of neuron - Potassium ions carry positive charge, so their increased movement out of the cell makes the inside even more negative than -70 mV. This increase in negativity is called hyperpolarization, and it makes the neuron less likely to fire.

Question: 4. Action potentials move along axons (a) More slowly in axons of large than in small diameter. (b) The direct action of acetylcholine on the axon membrane. (c) Activating the sodium-potassium pump at each point along the axon membrane. (d) More rapidly in myelinated than in non-myelinated axons.

Answer: (d) More rapidly in myelinated than in non-myelinated axons. - In myelinated axons the impulse jumps from one node of Ranvier to the next, which is much faster than continuous conduction along a non-myelinated axon. Larger diameter also increases speed, so option (a) is the reverse of the truth.

5. The surface on a neuron that discharges vesicles is the

(a) Dendrite

(b) Axon hillock

✓ (c) Presynaptic membrane

(d) Postsynaptic membrane.

Answer: (c) Presynaptic membrane - The presynaptic membrane of the axon terminal discharges synaptic vesicles containing neurotransmitter into the synaptic cleft. The postsynaptic membrane only receives the transmitter through its receptors.

6. An inhibitory postsynaptic potential occurs in a membrane made more permeable to

✓ (a) Potassium ions

(b) Sodium ions

(c) Calcium ions

(d) ATP

Answer: (a) Potassium ions - An inhibitory postsynaptic potential is produced when the membrane becomes more permeable to potassium ions (and chloride ions), so positive charge leaves the cell and the inside becomes more negative. This hyperpolarization moves the membrane away from threshold and inhibits firing.

7. The major inhibitory neurotransmitter of the brain is

(a) Acetylcholine

(b) Epinephrine

✓ (c) GABA

(d) Endorphin

Answer: (c) GABA - GABA (gamma-aminobutyric acid) is the main inhibitory neurotransmitter of the brain, reducing the activity of neurons and preventing over-excitation. Acetylcholine and epinephrine mainly act as excitatory transmitters at most of their sites.

8. Where are neurotransmitter receptors located

(a) On the nuclear membrane

(b) In the myelin sheet

(c) At nodes of Ranvier

✓ (d) On the postsynaptic membrane

Answer: (d) On the postsynaptic membrane - Neurotransmitter receptors are located on the postsynaptic membrane, facing the synaptic cleft. When transmitter released from the presynaptic side binds to them, ion channels open and a postsynaptic potential is produced.

9. Which selection is incorrectly paired?

(a) Forebrain -> Diencephalon

(b) Fore brain -> Cerebrum

(c) Midbrain -> Brainstem

✓ (d) Midbrain -> Cerebellum

Answer: (d) Midbrain -> Cerebellum - The cerebellum is a part of the hindbrain, not of the midbrain, so this pairing is incorrect. The diencephalon and cerebrum belong to the forebrain, and the midbrain forms the upper part of the brainstem.

10. Which of the following receptors is incorrectly paired with the type of energy is transduces?

(a) Mechanoreceptors -> Sound

✓ (b) Pain receptors -> Electricity

(c) Chemoreceptors -> Solute concentration

(d) Thermo receptors -> Heat

Answer: (b) Pain receptors -> Electricity - Pain receptors (nociceptors) respond to damaging mechanical, thermal or chemical stimuli, not to electricity as their specific energy, so this pairing is incorrect.

Mechanoreceptors do detect sound vibrations, chemoreceptors detect solute concentration and thermoreceptors detect heat.

## 2. Write short answers of the following questions:

**Question: 1. Why receptors are called transducers?**

Answer: A transducer is a device that converts one form of energy into another form. Sensory receptors convert the energy of a stimulus, such as light, sound, pressure or heat, into electrical energy in the form of nerve impulses. Because of this conversion of stimulus energy into action potentials, receptors are called biological transducers.

**Question: 2. Why neurolemma is polarized?**

Answer: At rest, the inside of the neuron contains more potassium ions and negatively charged proteins, while the outside has more sodium ions, and the sodium-potassium pump keeps pumping Na+ out and K+ in. As a result the inner surface of the neurolemma (axon membrane) is negative and the outer surface is positive, with a resting potential of about -70 mV. This separation of charge across the membrane is why the neurolemma is called polarized.

**Question: 3. What do you mean by threshold potential?**

Answer: Threshold potential is the minimum level of depolarization, about -55 mV, which a stimulus must produce in the neuron membrane for an action potential to be generated. A stimulus weaker than threshold produces no impulse, while any stimulus reaching threshold produces a full action potential. It therefore follows the all-or-none principle.

**Question: 4. Why impulses move faster in myelinated neurons?**

Answer: In myelinated neurons the myelin sheath insulates the axon, and the action potential is generated only at the exposed nodes of Ranvier. The impulse therefore jumps rapidly from node to node instead of traveling continuously along the whole membrane. This saltatory conduction makes conduction many times faster than in non-myelinated neurons.

Question: 5. Why interneurons are called association neuron?

Answer: Interneurons lie between sensory and motor neurons within the central nervous system and connect, or associate, them with each other. They receive information from sensory neurons, process and integrate it, and pass suitable commands to motor neurons. Because of this associating and connecting role, they are called association neurons.

Question: 6. What do you mean by saltatory conduction?

Answer: Saltatory conduction is the jumping movement of the nerve impulse from one node of Ranvier to the next along a myelinated axon. The action potential is regenerated only at the nodes, while the myelinated parts are insulated. This makes conduction fast and also saves energy for the neuron.

Question: 7. Differentiate between following (a) Electrical and Chemical synapse (b) CNS and PNS (c) Excitatory neurotransmitters and Inhibitory neurotransmitters (d) Parasympathetic nervous system and Sympathetic nervous system

Answer: (a) In an electrical synapse the impulse passes directly through gap junctions between cells, so transmission is very fast and can be two-way, while in a chemical synapse a neurotransmitter carries the signal

across the synaptic cleft, so transmission is slower and one-way. (b) The CNS consists of the brain and spinal cord and integrates information, while the PNS consists of cranial and spinal nerves and ganglia and carries impulses between the CNS and the body. (c) Excitatory neurotransmitters such as acetylcholine and glutamate depolarize the postsynaptic membrane and promote firing, while inhibitory ones such as GABA hyperpolarize it and prevent firing. (d) The sympathetic system prepares the body for emergency (fight or flight), increasing heartbeat and dilating pupils, while the parasympathetic system works during rest, slowing the heart and promoting digestion; the two act antagonistically to maintain balance.

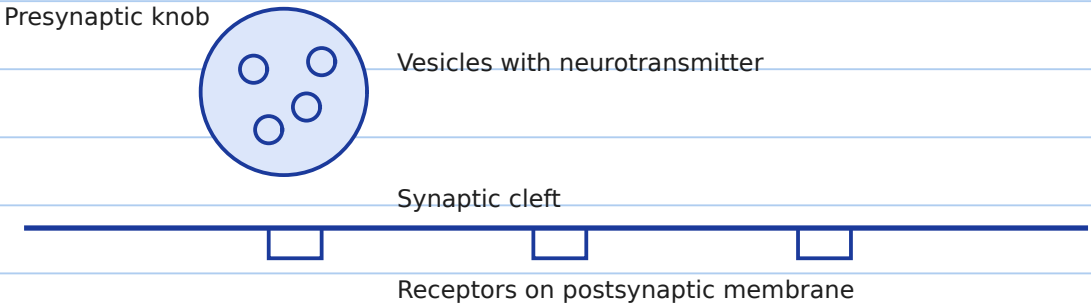
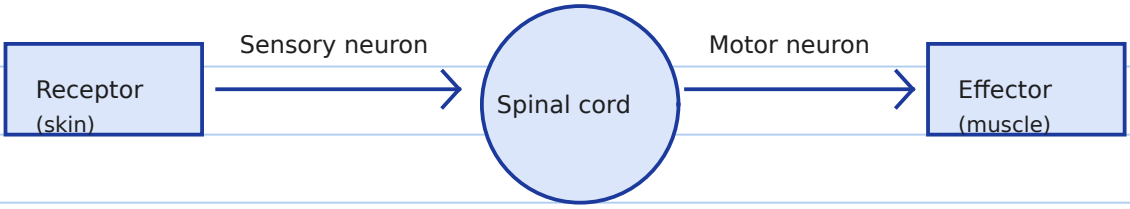



Fig.: Synapse (labelled diagram)

Question: 8. Define following terms (a) Endorphins (b) Pacinian corpuscles (c) Nociceptors (d) Reflex arc (e) Stroke (f) Meningitis

Answer: (a) Endorphins are natural pain-relieving chemicals produced by the brain that block pain pathways and produce a feeling of well-being. (b) Pacinian corpuscles are pressure receptors deep in the skin and tissues that detect deep pressure and vibrations. (c) Nociceptors are pain receptors that respond to damaging stimuli and produce the sensation of pain. (d) A reflex arc is the pathway of a reflex action: receptor, sensory neuron, interneuron in the CNS, motor neuron and effector. (e) Stroke is the death of brain tissue due to blockage or bursting of a blood vessel supplying the brain. (f) Meningitis is the inflammation of the meninges, the protective membranes around the brain and spinal cord, usually caused by bacterial or viral infection.




Path: receptor -> sensory neuron -> spinal cord -> motor neuron -> effector

Fig.: Reflex (labelled diagram)

## 3. Write detailed answers of the following questions:

Question: 1. Describe structure and function of neurons.

Answer: A neuron consists of three main parts: the cell body (soma), which contains the nucleus and cytoplasm and controls the metabolic activities of the cell; the dendrites, which are short branched processes that receive impulses and carry them towards the cell body; and the axon, a long fiber that carries impulses away from the cell body to other neurons, muscles or glands, often covered by a myelin sheath with nodes of Ranvier. Functionally, neurons generate and conduct nerve impulses, so sensory neurons carry information from receptors to the CNS, interneurons integrate it, and motor neurons carry commands to effectors. In this way neurons make rapid communication and coordination in the body possible.

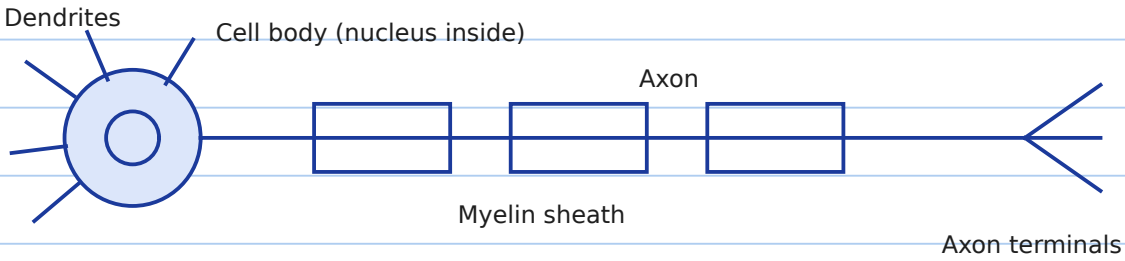



Fig.: Neuron (labelled diagram)

Question: 2. Describe mechanism of action membrane potential.

Answer: The action potential is a rapid, temporary reversal of membrane potential produced when a stimulus reaches threshold. First, voltage-gated sodium channels open and Na+ rushes into the axon, so the inside becomes positive (up to about +40 mV); this is depolarization. Then sodium channels close and potassium channels open, K+ moves out, and the membrane returns to its negative resting value; this is repolarization,

sometimes followed by brief hyperpolarization. The sodium-potassium pump then restores the original ion distribution, and the action potential travels along the axon as each neighboring region is depolarized in turn.

Question: 3. Explain vascular and infectious disorders of CNS.

Answer: Vascular disorders of the CNS are caused by disturbed blood supply to the brain, the most important being stroke, in which a clot blocks a cerebral vessel or a vessel bursts and bleeds, killing the brain tissue deprived of oxygen and causing paralysis, loss of speech or death. Infectious disorders are caused by bacteria, viruses or other pathogens that invade the CNS, such as meningitis, the inflammation of the meninges, and encephalitis, the inflammation of the brain itself. These infections cause fever, severe headache, neck stiffness and damage to nervous tissue, and need prompt medical treatment.

Question: 4. Describe anatomy and physiology of human brain.

Answer: The human brain is protected inside the skull by the bony cranium, three meninges and cerebrospinal fluid, and is divided into forebrain, midbrain and hindbrain. The forebrain includes the cerebrum, the largest part, whose cerebral cortex controls intelligence, memory, voluntary actions and sensations, and the diencephalon (thalamus and hypothalamus), which relays sensory information and regulates hunger, thirst, body temperature and endocrine activity. The small midbrain acts mainly as a relay center for visual and auditory information, while the hindbrain consists of the cerebellum, which coordinates movement and balance, and the pons and medulla, which control vital involuntary functions such as breathing, heartbeat and blood pressure.

Question: 5. Describe sensory receptors and their working.

Answer: Sensory receptors are specialized cells or neuron endings that detect a specific stimulus and convert it into nerve impulses. They are classified by the stimulus they detect: mechanoreceptors for touch, pressure and sound, chemoreceptors for taste, smell and blood chemicals, thermoreceptors for temperature, photoreceptors (rods and cones) for light, and nociceptors for pain. When a stimulus of sufficient strength acts on a receptor, it produces a generator potential, and if this reaches threshold, action potentials are fired whose frequency reflects the strength of the stimulus. These impulses travel through sensory neurons to the CNS, where the sensation is perceived and a suitable response is organized.

# Chapter 18: Chemical Coordination

## 1. Encircle the correct choice.

1. Which of the following statements about hormones is incorrect?

(a) They are produced by endocrine glands.

(b) They are modified amino acids, peptides or steroid molecules.

(c) They are carried by the circulatory system.

✓ (d) They are used to communicate between different organisms.

Answer: (d) They are used to communicate between different organisms. - Hormones are chemical messengers that coordinate activities within the same organism, being carried by the blood from endocrine glands to their target organs. Communication between different organisms is done by pheromones, not by hormones.

2. Oxytocin and ADH are produced by the _____ and stored in the _____.

✓ (a) Hypothalamus; Neurohypophysis

(b) Adenohypophysis; Kidneys

(c) Adrenal cortex; Adrenal medulla

(d) Posterior pituitary; Anterior pituitary

Answer: (a) Hypothalamus; Neurohypophysis - Oxytocin and ADH (vasopressin) are actually synthesized by neurosecretory cells of the hypothalamus. They are then transported down the axons and stored in the neurohypophysis (posterior pituitary), from where they are released into the blood.

3. Tropic hormones from the anterior pituitary directly affect the release of which of the following?

(a) Parathyroid hormone

(b) Calcitonin

(c) Epinephrine

✓ (d) Thyroxine

Answer: (d) Thyroxine - The anterior pituitary secretes thyroid stimulating hormone (TSH), a tropic hormone, which directly stimulates the thyroid gland to release thyroxine. Parathyroid hormone, calcitonin and epinephrine are not controlled by pituitary tropic hormones in this way.

4. Which of the following endocrine disorders is /are not correctly matched with the malfunctioning gland?

(a) Diabetes -> Pancreas

(b) Giantism -> Pituitary

✓ (c) Goiter -> Adrenal medulla

(d) Tetany -> Parathyroid

Answer: (c) Goiter -> Adrenal medulla - Goiter is the enlargement of the thyroid gland, usually due to iodine deficiency, and has nothing to do with the adrenal medulla. The other pairs are correctly matched: diabetes with pancreas, giantism with pituitary and tetany with parathyroid.

5. Which hormone exerts antagonistic action to PTH?

(a) Thyroxin

✓ (b) Calcitonin

(c) Growth hormone

(d) Epinephrine

Answer: (b) Calcitonin - Parathyroid hormone (PTH) raises blood calcium by releasing calcium from bones, while calcitonin from the thyroid lowers blood calcium by promoting its deposition in bones. The two hormones

thus act antagonistically to keep blood calcium constant.

6. Which of the following are synthesized from the amino acid tyrosine?

(a) Epinephrine

(b) Catecholamines

(c) Thyroxin

✓ (d) All of the following

Answer: (d) All of the following - Epinephrine and the other catecholamines of the adrenal medulla are derived from the amino acid tyrosine, and thyroxine of the thyroid gland is also formed from tyrosine units. Hence all the listed hormones share tyrosine as their starting material.

7. If the adrenal cortex were removed, which group of hormones would be most affected?

✓ (a) Steroid

(b) Peptide

(c) Tropic

(d) Amino acid derived

Answer: (a) Steroid - The adrenal cortex produces steroid hormones, namely glucocorticoids, mineralocorticoids and sex steroids, all made from cholesterol. Removal of the cortex would therefore directly abolish this steroid group, while peptide and amino-acid-derived hormones are made elsewhere.

8. Which of the following hormones is (are) secreted by the adrenal gland in response to stress and promote(s) the synthesis of glucose from non-carbohydrate substrates?

(a) Glucagon

(b) Glucocorticoids

(c) Epinephrine

(d) Thyroxin

Answer: (b) Glucocorticoids - Glucocorticoids such as cortisol are secreted by the adrenal cortex during stress and promote gluconeogenesis, the synthesis of glucose from amino acids and other non-carbohydrate substrates. This raises blood glucose to meet the increased energy demand of the stress condition.

9. Which of the following stimulates the contraction of uterine muscle?

(a) Oxytocin

(b) Thyroxin

(c) Growth hormone

(d) Insulin

Answer: (a) Oxytocin - Oxytocin, released from the posterior pituitary, strongly stimulates contraction of the smooth muscle of the uterus during childbirth. It also causes contraction of mammary glands for the ejection of milk after birth.

10. A distinctive feature of the mechanism of action of thyroid hormones and steroid hormones is that

(a) These hormones are regulated by feedback loop

(b) These hormones bind with specific receptor protein on the plasma membrane of target cells.

(c) These hormones bind to receptors inside cells

(d) Target cells react more rapidly to these hormones than to local regulators.

Answer: (c) These hormones bind to receptors inside cells - Thyroid and

steroid hormones are lipid soluble, so they pass through the plasma membrane and bind to receptors inside the cell, in the cytoplasm or nucleus. The hormone-receptor complex then switches specific genes on or off, changing protein synthesis in the target cell.

## 2. Write short answers of the following questions:

**Question: 1. Why hydrophilic hormones need secondary messengers?**

Answer: Hydrophilic (water-soluble) hormones such as peptides and catecholamines cannot pass through the lipid plasma membrane of the target cell. They therefore bind to receptors on the cell surface, and the message is carried inside the cell by a second messenger such as cyclic AMP, which activates enzymes and produces the response. The hormone itself acts as the first messenger outside the cell.

**Question: 2. Why hormones are called chemical messengers?**

Answer: Hormones are chemical substances produced by endocrine glands in very small amounts and carried by the blood to distant target organs. Like messengers, they carry information and instructions from one part of the body to another and tell the target cells to start, stop or adjust a particular activity. For this reason they are called chemical messengers.

**Question: 3. Why posterior pituitary is called neurohypophysis?**

Answer: The posterior pituitary is made of nervous tissue and is actually an extension of the hypothalamus, connected to it by nerve fibers. It does not manufacture its own hormones; it only stores and releases oxytocin and ADH made by hypothalamic neurons. Because of this nervous origin and structure, it is called the neurohypophysis, unlike the glandular anterior pituitary (adenohypophysis).

Question: 4. Enlist hormones of anterior pituitary?

Answer: The anterior pituitary secretes six main hormones: growth hormone (GH or somatotropin) for body growth, thyroid stimulating hormone (TSH) for the thyroid, adrenocorticotropic hormone (ACTH) for the adrenal cortex, follicle stimulating hormone (FSH) and luteinizing hormone (LH) for the gonads, and prolactin for milk production in the mammary glands. Most of these, except GH and prolactin, are tropic hormones that control other endocrine glands.

Question: 5. Why pancreas is called as exocrine and endocrine gland?

Answer: The pancreas works in both ways, so it is a mixed gland. Its acinar cells act as an exocrine part and pour pancreatic juice containing digestive enzymes through a duct into the duodenum. Its islets of Langerhans act as the endocrine part and release hormones directly into the blood: insulin from beta cells, glucagon from alpha cells and somatostatin from delta cells.

Question: 6. Why secretions of adrenal cortex are called corticosteroids?

Answer: All hormones of the adrenal cortex are chemically steroids and are synthesized from cholesterol. Because they are steroid hormones produced by the cortex, they are collectively called corticosteroids. They include glucocorticoids (e.g. cortisol), mineralocorticoids (e.g. aldosterone) and small amounts of sex steroids.

Question: 7. Differentiate between following (a) Hypothyroidism and Hyperthyroidism (b) Insulin and glucagon (c) Calcitonin and Parathyroid hormone (d) Lipophilic and hydrophilic hormones

Answer: (a) Hypothyroidism is under-secretion of thyroxine, causing low

metabolic rate, weight gain and sluggishness (cretinism in children, myxedema in adults), while hyperthyroidism is over-secretion, causing high metabolic rate, weight loss, rapid heartbeat and nervousness. (b) Insulin, from beta cells of the pancreas, lowers blood glucose by promoting its uptake and glycogen formation, while glucagon, from alpha cells, raises blood glucose by breaking down glycogen in the liver. (c) Calcitonin, from the thyroid, lowers blood calcium by depositing it in bones, while parathyroid hormone raises blood calcium by releasing it from bones and increasing its absorption. (d) Lipophilic hormones are lipid soluble (steroids, thyroxine), pass through the cell membrane and bind to internal receptors, while hydrophilic hormones are water soluble (peptides, catecholamines), cannot cross the membrane and act through surface receptors and second messengers.

Question: 8. Define following terms (a) Hormones (b) Goiter (c) Thymosin (d) GnTH (e) Atrial natriuretic hormone (f) Erythropoietin

Answer: (a) Hormones are chemical messengers secreted by endocrine glands directly into the blood, which regulate the activity of specific target organs. (b) Goiter is the enlargement of the thyroid gland, usually caused by iodine deficiency, seen as a swelling in the neck. (c) Thymosin is the hormone of the thymus gland that promotes the maturation of T-lymphocytes for immunity. (d) GnTH (gonadotropic hormones) are FSH and LH from the anterior pituitary that stimulate the gonads to produce gametes and sex hormones. (e) Atrial natriuretic hormone is released by the heart atria when blood pressure/volume rises, and it increases excretion of sodium and water by the kidneys to lower blood pressure. (f) Erythropoietin is a hormone from the kidneys that stimulates the bone marrow to produce red blood cells when blood oxygen is low.

## 3. Write detailed answers of the following questions:

**Question: 1. Define hormone? Describes hormone action with the help of suitable diagram.**

Answer: A hormone is a chemical messenger produced by an endocrine gland, released directly into the blood in small quantity, and carried to a specific target organ where it regulates its activity. Hydrophilic hormones cannot cross the cell membrane, so they bind to surface receptors and act through a second messenger such as cyclic AMP, which activates enzymes inside the cell and produces a rapid response. Lipophilic hormones such as steroids and thyroxine pass through the membrane, bind to receptors inside the cell or nucleus, and switch on specific genes, producing a slower but longer-lasting response through new protein synthesis. In both modes the hormone is recognized only by cells carrying its specific receptor, which is why its action remains target-specific.

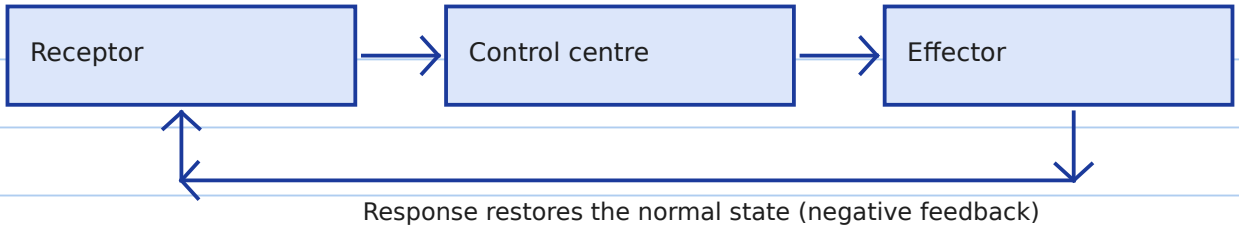



Fig.: Feedback (labelled diagram)

**Question: 2. Explain pituitary gland their lobes and hormones of pituitary glands.**

Answer: The pituitary gland is a pea-sized master gland hanging from the hypothalamus below the brain, and it has two main lobes. The anterior lobe (adenohypophysis) is glandular and secretes growth hormone, prolactin and the tropic hormones TSH, ACTH, FSH and LH, which control growth, milk production and the thyroid, adrenal cortex and gonads respectively. The posterior lobe (neurohypophysis) is nervous in origin; it stores and releases oxytocin, which contracts the uterus and ejects milk, and ADH, which promotes reabsorption of water in the kidneys. Because it controls so many other glands through its tropic hormones, the pituitary is called the master endocrine gland.

Question: 3. Explain position and physiology of adrenal gland.

Answer: There are two adrenal (suprarenal) glands, one sitting like a cap on top of each kidney. Each gland has an outer cortex and an inner medulla with different functions: the cortex secretes corticosteroids - glucocorticoids such as cortisol, which regulate carbohydrate metabolism and stress responses, mineralocorticoids such as aldosterone, which regulate salt, water and blood pressure, and small amounts of sex hormones. The medulla secretes the catecholamines epinephrine and norepinephrine, which prepare the body for emergency by increasing heartbeat, blood pressure and blood glucose. Together the two parts help the body handle stress and maintain salt, water and energy balance.

Question: 4. Describe hormones and their functions of other endocrine tissues.

Answer: Besides the main glands, several other tissues produce hormones with important functions. The gastrointestinal tract secretes gastrin, secretin and cholecystokinin, which coordinate digestion by controlling gastric juice, pancreatic juice and bile release; the kidneys secrete erythropoietin for red blood cell formation and renin for blood pressure regulation; and the heart releases atrial natriuretic hormone, which lowers blood pressure by increasing salt and water excretion. The placenta during pregnancy secretes HCG and progesterone to maintain pregnancy, the thymus secretes thymosin for T-lymphocyte maturation, and the pineal gland secretes melatonin, which regulates the sleep-wake cycle.

Question: 5. Describe mechanism of positive feedback.

Answer: In positive feedback, the result of a process further increases the same process instead of stopping it, so the response is amplified until the

task is completed. A clear example is childbirth: contraction of the uterus pushes the baby against the cervix, stretching it, and this stretching signal causes release of more oxytocin, which makes contractions still stronger, until the baby is delivered and the stimulus ends. Another example is blood clotting, where activated clotting factors activate more and more factors until the clot seals the wound.

# Chapter 19: Animal Behaviour

## 1. Encircle the correct choice.

1. i) The decrease in response to repeated or continuous stimulation is called

(a) Insight

(b) Habituation ✓

(c) Maturation

(d) Instinct

Answer: (b) Habituation — When a stimulus is repeated again and again without any reward or harm, the animal gradually stops responding to it. This simplest form of learning is called habituation.

2. ii) The cyclic physiological patterns of activities in an organism that are in response to periodic environmental changes is called

(a) Biological Clock

(b) Biological rhythms

(c) Circadian rhythms

(d) All of Above ✓

Answer: (d) All of Above — Regular cycles of activity controlled by an internal biological clock in response to periodic changes such as day and night are called biological rhythms, and the daily ones are circadian rhythms. All the given terms are therefore related to the same phenomenon.

3. iii) A directional movement of an organism toward or away from the particular stimulus is called

(a) Kinesis

(b) Taxis

(c) Reflex

(d) Fixed Action Pattern

Answer: (b) Taxis — Taxis is a directed movement in which the whole organism moves toward a stimulus (positive taxis) or away from it (negative taxis). In kinesis, by contrast, the response is not directed toward or away from the stimulus.

4. iv) Simplest form of learning, in which the animals learn not to respond or ignore to a repeated, irrelevant stimulus.

(a) Imprinting

(b) Kinesis

(c) Habituation

(d) Fixed Action Pattern

Answer: (c) Habituation — Habituation is the simplest form of learning in which an animal learns to ignore a repeated stimulus that brings neither benefit nor harm. It saves the animal from wasting energy on useless responses.

5. v) Which one of the following is not an instinct behaviour?

(a) Migration of Salmon Fish

(b) Dances of bees

(c) Territorial behaviour in Gorillas

(d) Nest building by birds

Answer: (c) Territorial behaviour in Gorillas — Migration of salmon, dances of bees and nest building by birds are classic inborn instinctive patterns performed without learning. Territorial defence in gorillas is classified as

agonistic social behaviour shown in interaction with other members of the group.

6. vi) Wolfgang Kohler in 1920s performed the experiment on the behaviour of

(a) Rats

✓ (b) Chimpanzees

(c) Graylag Goose

(d) Dog

Answer: (b) Chimpanzees — In the 1920s Wolfgang Kohler experimented on chimpanzees and showed insight learning. The chimpanzees suddenly solved problems, for example by joining sticks or piling boxes to reach bananas, without gradual trial and error.

7. vii) Workers (honey bee) are

(a) Fertile Females

(b) Fertile Male

✓ (c) Sterile Female

(d) Sterile Male

Answer: (c) Sterile Female — In a honey bee colony the workers are females that cannot reproduce. They perform all the colony work such as collecting food, cleaning, feeding larvae and defending the hive, while the queen is the fertile female.

8. viii) Agonistic behaviour is the type of

(a) Innate Behaviour

(b) Learning behaviour

✓ (c) Social Behaviour

(d) Instinctive behaviour

Answer: (c) Social Behaviour — Agonistic behaviour includes aggression, threat displays, submission and fighting between members of the same species. Because it occurs during social interactions over food, mates or territory, it is a type of social behaviour.

9. ix) Migration of Salmon Fish is the example of

(a) Orientation behaviour

(b) Learning behaviour

(c) Social Behaviour

✓ (d) Instinctive behaviour

Answer: (d) Instinctive behaviour — Salmon migrate from the sea back to the river of their birth to breed without ever being taught the route. This inborn, inherited pattern is an example of instinctive behaviour.

10. x) Skinner box was used for

(a) Classic Conditioning

✓ (b) Operant Learning

(c) Latent Learning

(d) Insight Learning

Answer: (b) Operant Learning — B. F. Skinner used the Skinner box to study operant learning in rats and pigeons. The animal learned to press a lever to obtain food as a reward, so its behaviour was shaped by its consequences.

## 2. Write short answer of the following questions:

Question: 11. i) Define stimuli. How organisms respond to different stimuli?

Answer: A stimulus is any detectable change, physical or chemical, in the internal or external environment of an organism, such as light, heat, sound, touch or a chemical. Organisms detect stimuli through receptors and respond through nerve impulses, hormones and muscles or glands. The response may be movement toward or away from the stimulus, a reflex, or a change in internal activity that helps survival.

Question: 12. ii) What is reflex? Give three examples of reflexes in vertebrates.

Answer: A reflex is a rapid, automatic and involuntary response to a stimulus that does not require conscious thought and follows a fixed reflex arc of receptor, sensory neuron, central nervous system, motor neuron and effector. Examples in vertebrates are the knee-jerk reflex, blinking of the eye when an object approaches, and withdrawal of the hand from a hot object. Salivation at the sight or smell of food is another common example.

Question: 13. iii) What is the relationship between heredity and behaviour?

Answer: Heredity provides the basic pattern of behaviour through genes, so innate behaviours such as reflexes, instincts and fixed action patterns are inherited and appear without learning. Genes also set the limits within which learning can modify behaviour. Thus behaviour is a product of both inherited tendencies and experience, and closely related animals show similar behaviour because they share similar genes.

Question: 14. iv) Define biological rhythms. How are biological rhythms important to man?

Answer: Biological rhythms are the cyclic physiological patterns of activity

in an organism that occur in response to periodic environmental changes and are controlled by an internal biological clock. In man they regulate the sleep-wake cycle, body temperature, hormone secretion and alertness in a roughly 24-hour circadian rhythm. Disturbance of these rhythms, for example by night shifts or jet lag, causes tiredness and poor performance, so they are important for health and normal daily life.

Question: 15. v) What do you know about taxis? Give examples of positive taxis and negative taxis.

Answer: Taxis is a directional movement of a whole organism toward or away from a particular stimulus such as light, heat or a chemical. Movement toward the stimulus is positive taxis, for example moths flying toward light (positive phototaxis) and a sperm moving toward chemicals from the egg (positive chemotaxis). Movement away from the stimulus is negative taxis, for example a cockroach running away from light and an earthworm moving away from strong light or dryness.

Question: 16. vi) What is the difference between innate behaviour and learning behaviour?

Answer: Innate behaviour is inborn and inherited, is present at birth, needs no experience or practice, and is similar in all members of a species, for example reflexes and nest building. Learning behaviour is acquired during life through experience and practice, differs from individual to individual, and can be changed or improved. Innate behaviour is controlled mainly by genes, while learned behaviour depends on the nervous system and memory.

Question: 17. vii) How do bees communicate about food sources?

Answer: Honey bees communicate the distance and direction of a food source by performing dances on the comb, as described by Karl von Frisch. A round dance shows that food is near the hive, while a waggle dance, performed in a figure-of-eight with a waggling run, shows a distant source. The angle of the waggle run to the vertical indicates direction with respect to the sun, and the duration of waggling indicates distance.

Question: 18. viii) How do old world monkeys defend their territory?

Answer: Old world monkeys live in troops and defend their home territory by group displays and aggression against intruding troops. Dominant males give loud calls, show threatening gestures and expose their large canine teeth, and may chase or fight intruders. Such agonistic displays usually settle the dispute without serious injury and keep other troops away from their food and resting sites.

## 3. Write detailed answers of the following questions

Question: 19. i) Define innate behaviour in term of reflexes shown by vertebrates and invertebrates

Answer: Innate behaviour is the inherited behaviour performed by an organism from birth in response to a stimulus without prior experience or learning. Its simplest form is the reflex, a rapid automatic response through a reflex arc, such as the knee-jerk, blinking and hand-withdrawal reflexes in vertebrates and the withdrawal of a snail or earthworm on touch in invertebrates. In invertebrates, innate behaviour also appears as kinesis and taxis, for example woodlice moving faster in dry conditions until they reach moisture. In vertebrates, more complex innate patterns develop into instincts and fixed action patterns, but all are built on inherited nervous pathways and need no practice.

Question: 20. ii) Describe the construction of an intricate web by a spider as instinct behaviour.

Answer: Web building by a spider is a classic example of instinct, because a young spider raised in isolation builds a perfect web of its species without ever seeing one or being taught. The spider first lays a bridge thread across a gap, then builds a frame and radial threads from a central hub. It then lays a temporary spiral and replaces it with a sticky capture spiral, testing tension and spacing as it works. The whole sequence is a fixed, inherited pattern released by internal readiness and simple stimuli, and it serves the adaptive purpose of catching prey.

Question: 21. iii) What is imprinting? Describe the imprinting in young ducks. How is it adaptive?

Answer: Imprinting is a rapid type of learning that occurs during a short sensitive period early in life, in which a young animal fixes its attention on the first moving object it sees and thereafter follows it. Konrad Lorenz showed that newly hatched ducks and goslings follow the first moving object, normally their mother, but they can even be imprinted on a human or an object if that is what they first encounter. It is adaptive because the young stay close to the mother, who protects, warms and guides them to food, greatly increasing their chances of survival.

Question: 22. iv) Describe the agonistic behaviour and relate it with the maintenance of social order in terms of territories and dominance hierarchies.

Answer: Agonistic behaviour is the aggressive and submissive behaviour shown when animals of the same species compete for food, mates, shelter or territory; it includes threat displays, posturing, vocalizations, chasing

and sometimes fighting. By defending a territory, an animal secures enough food and a breeding place, and rivals are kept out mostly by displays rather than injurious fights. Within a group, contests establish a dominance hierarchy or pecking order in which higher-ranking animals get first access to resources and lower-ranking ones submit. In this way agonistic behaviour maintains social order, reduces repeated fighting and stabilizes group life.

# Chapter 20: Reproduction

## 1. Encircle the correct choice.

1. i) Ovarian cycle includes changes in ovarian follicles to secondary follicle and finally to

(a) Graafian follicle

(b) Primary follicle

(c) Vesicular follicle

✓ (d) Both "a" and "c"

Answer: (d) Both "a" and "c" — During the ovarian cycle a primary follicle develops into a secondary follicle and then into the mature follicle. This mature follicle is called both the Graafian follicle and the vesicular follicle, so both names are correct.

2. ii) In males, FSH promotes the development and maturation of?

(a) Testosterone

✓ (b) Sperm cells

(c) Egg cells

(d) Polar cells

Answer: (b) Sperm cells — In males, follicle stimulating hormone (FSH) from the pituitary acts on the seminiferous tubules and Sertoli cells of the testes. It promotes spermatogenesis, that is, the development and maturation of sperm cells.

3. iii) The menstrual cycle is primarily regulated by all of the following except:

(a) FSH

(b) Estrogen

✓ (c) Androgen

(d) Progesterone

Answer: (c) Androgen — The menstrual cycle is regulated by FSH and LH from the pituitary together with estrogen and progesterone from the ovary. Androgens are male hormones and do not primarily regulate the female menstrual cycle.

4. iv) FSH induces the Graafian follicle to produce

(a) LH

✓ (b) Estrogen

(c) Androgen

(d) Progesterone

Answer: (b) Estrogen — Under the influence of FSH the growing Graafian follicle secretes estrogen. Estrogen causes growth and repair of the uterine lining during the first half of the menstrual cycle.

5. v) Key hormones in female reproductive system that regulates menstrual cycle. I. Estrogen II. Progesterone III. Testosterone

(a) I only

(b) II only

✓ (c) I and II

(d) III only

Answer: (c) I and II — Estrogen and progesterone are the two key ovarian hormones that regulate the menstrual cycle, controlling follicle growth, ovulation and preparation of the uterus. Testosterone is the main male hormone and is not a key regulator of the cycle.

6. vi) What is the medical term for a miscarriage?

✓ (a) Spontaneous abortion

(b) Induced abortion

(c) Ectopic pregnancy

(d) Both "a" and "b"

Answer: (a) Spontaneous abortion — A miscarriage is the natural, spontaneous loss of a pregnancy before the foetus can survive outside the uterus. In medical terms it is called spontaneous abortion, whereas induced abortion is a deliberate termination.

7. vii) Syphilis is a sexually transmitted disease caused by

✓ (a) Bacteria

(b) Virus

(c) Protozoan

(d) Yeast

Answer: (a) Bacteria — Syphilis is caused by the bacterium Treponema pallidum, transmitted through sexual contact. It produces sores and rashes and, if untreated, can damage the heart and nervous system, but it can be cured with antibiotics.

8. viii) Structure that protects the testes and keeps them at a temperature several degrees below the normal body temperature.

(a) Epididymis

✓ (b) Scrotum

(c) Penis

(d) All of the above

Answer: (b) Scrotum — The scrotum is the sac of skin that holds and protects the testes outside the body cavity. It keeps them about 2-3

degrees Celsius below normal body temperature, which is necessary for normal sperm production.

9. ix) In-vitro fertilization (IVF) is a method used to:

✓ (a) Treat infertility

(b) Enhance hormones

(c) Prevent STDs

(d) None of the above

Answer: (a) Treat infertility — In IVF, eggs are fertilized by sperm outside the body in the laboratory and the resulting embryo is placed in the uterus. It is an assisted reproductive technique used to help couples who cannot conceive naturally.

10. x) Which of the following is a potential cause of female infertility?

(b) Endometriosis

(c) Fallopian tube blockage

✓ (d) All of these

Answer: (d) All of these — PCOS disturbs ovulation, endometriosis damages the reproductive organs, and blocked fallopian tubes prevent the egg and sperm from meeting. Any of these can cause female infertility, so all of these are correct.

2. Write short answer of the following questions:

Question: 11. i) Why urethra in male is called urinogenital duct?

Answer: In the male, the urethra is a common passage for both urine and semen. Urine from the bladder and semen from the vas deferens both leave the body through it, though never at the same time. Because it serves both the urinary and the genital systems, it is called the

urinogenital duct.

Question: 12. ii) Why hormonal system of female is better than male?

Answer: The female hormonal system is considered better, or more finely regulated, because it works in a precise cyclic manner with feedback control between the pituitary and the ovaries. Estrogen and progesterone rise and fall in a coordinated sequence that controls follicle growth, ovulation, preparation of the uterus and, if pregnancy occurs, its maintenance. This regular hormonal cycle prepares the body every month for a possible pregnancy, a level of cyclic coordination not found in the male.

Question: 13. iii) Why testes descend down in human male foetus before birth?

Answer: The testes develop inside the abdomen of the foetus but descend into the scrotum before birth because sperm production requires a temperature about 2-3 degrees Celsius lower than the internal body temperature. Inside the abdomen the heat would damage the sperm-forming cells and cause sterility. Descent into the cooler scrotum therefore ensures normal spermatogenesis after puberty.

Question: 14. iv) What are symptoms of gonorrhea?

Answer: Gonorrhea is a bacterial sexually transmitted disease caused by Neisseria gonorrhoeae. Its symptoms include a burning sensation and pain during urination and a thick yellow-green discharge from the urethra or vagina. If untreated it can inflame the reproductive organs and cause infertility, and some infected persons, especially women, may show few early symptoms.

Question: 15. v) Why inner uterine wall thicken during secretory phase?

Answer: During the secretory phase, progesterone from the corpus luteum makes the inner uterine wall (endometrium) thicken, become rich in blood vessels and glands that secrete nutrients. This prepares a soft, nourishing bed in which a fertilized egg can implant and grow. If fertilization does not occur, progesterone falls and this thickened lining is shed during menstruation.

Question: 16. vi) Enlist causes of male and female infertility.

Answer: Causes of male infertility include low sperm count, poor sperm motility or abnormal sperm, blocked vas deferens, hormonal imbalance, undescended testes and infections. Causes of female infertility include ovulation disorders such as PCOS, blocked fallopian tubes, endometriosis, uterine abnormalities and hormonal imbalance. Age, smoking and general ill health can contribute to infertility in both sexes.

Question: 17. vii) Differentiate between the following (a) Testes and Ovaries (b) Spermatogenesis and Oogenesis

Answer: (a) Testes are the male gonads, located in the scrotum, which produce sperm and testosterone, while ovaries are the female gonads, located in the pelvic cavity, which produce eggs (ova) and estrogen and progesterone. (b) Spermatogenesis is the formation of sperm in the testes; it occurs continuously from puberty and produces four functional sperm from each primary spermatocyte. Oogenesis is the formation of eggs in the ovaries; it begins before birth, occurs cyclically, and produces only one functional ovum and polar bodies from each primary oocyte.

## 3. Write detailed answers to the following questions.

Question: 18. i) Describe the structures of male reproductive system identifying their functions.

Answer: The male reproductive system consists of the testes, epididymis, vas deferens, accessory glands and penis. The testes, held in the scrotum, contain seminiferous tubules where sperm are produced and cells that secrete testosterone. Sperm mature and are stored in the coiled epididymis, then pass through the vas deferens to the urethra. The seminal vesicles, prostate gland and Cowper glands add fluids that nourish and protect sperm and form semen. The penis is the copulatory organ that delivers semen into the female reproductive tract during intercourse.

Question: 19. ii) Explain the principal reproductive hormones of human male and explain their role in the maintenance and functioning of reproductive system

Answer: The principal male reproductive hormones are gonadotropin-releasing hormone from the hypothalamus, LH and FSH from the pituitary, and testosterone from the testes. LH stimulates the interstitial (Leydig) cells of the testes to secrete testosterone, which is responsible for sperm production and the development and maintenance of male secondary sexual characters such as beard, deep voice and body hair. FSH acts on the seminiferous tubules and Sertoli cells to promote spermatogenesis. Through negative feedback, testosterone keeps the pituitary hormones at a steady level, maintaining normal functioning of the system.

Question: 20. iii) Explain the structures of female reproductive system and describe their functions.

Answer: The female reproductive system consists of the ovaries, fallopian

tubes, uterus, cervix and vagina. The ovaries produce eggs and the hormones estrogen and progesterone, and release one egg each month at ovulation. The fallopian tubes (oviducts) carry the egg toward the uterus and are the usual site of fertilization. The uterus is a muscular organ whose lining, the endometrium, receives the implanted embryo and nourishes the developing foetus during pregnancy. The cervix is the narrow lower end of the uterus opening into the vagina, which receives sperm during intercourse and forms the birth canal during childbirth.

Question: 21. iv) Describe phases of menstrual cycle.

Answer: The menstrual cycle of about 28 days has three main phases. In the menstrual phase (days 1-5), the thickened endometrium breaks down and is shed with blood through the vagina because no pregnancy has occurred. In the proliferative or follicular phase (days 6-14), FSH develops an ovarian follicle, and estrogen from the follicle repairs and thickens the endometrium, ending with ovulation around day 14. In the secretory or luteal phase (days 15-28), the corpus luteum secretes progesterone, which makes the endometrium thick, vascular and secretory for possible implantation; if fertilization fails, hormone levels fall and the cycle repeats.

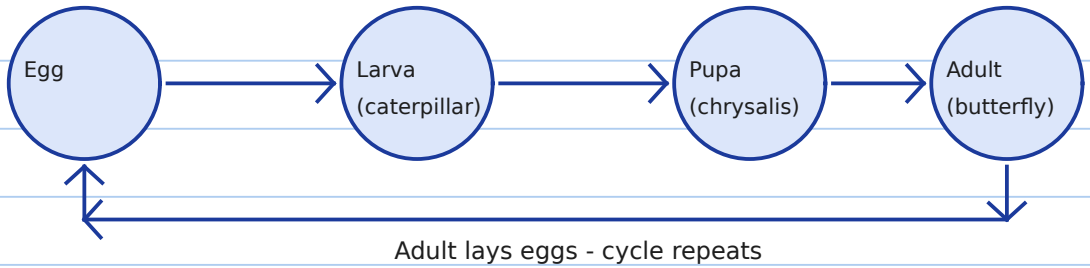



Fig.: Lifecycle (labelled diagram)

# Chapter 21: Development and Aging

## 1. Encircle the correct choice

1. i) The genetic material of a fertilized egg is a combination of

(a) The mother's DNA only

(b) The father's DNA only

✓ (c) Both the mother's and father's DNA

(d) None of the above

Answer: (c) Both the mother's and father's DNA — At fertilization the haploid nucleus of the sperm fuses with the haploid nucleus of the egg. The resulting zygote therefore contains a complete diploid set of DNA, half from the mother and half from the father.

2. ii) Human embryonic development occurs in the

✓ (a) Uterus

(b) Fallopian tube

(c) Cervix

(d) Vagina

Answer: (a) Uterus — After fertilization in the fallopian tube, the embryo implants in the wall of the uterus. It then develops there, nourished through the placenta, until birth.

3. iii) The process of cell differentiation involves:

(a) The formation of gametes

(b) The formation of zygotes

✓ (c) The specialization of cells for different functions

(d) The replication of DNA

Answer: (c) The specialization of cells for different functions — In differentiation, cells that have the same genes switch on different genes and become specialized in structure and function, for example into muscle, nerve or blood cells. This specialization builds the different tissues and organs of the body.

4. iv) The implantation of the embryo in the uterus occurs at

✓ (a) 1 week after fertilization

(b) 2 weeks after fertilization

(c) 4 weeks after fertilization

(d) 8 weeks after fertilization

Answer: (a) 1 week after fertilization — The fertilized egg divides during its journey down the fallopian tube and forms a blastocyst. About six to seven days, roughly one week, after fertilization the blastocyst implants in the endometrium of the uterus.

5. v) Aging is characterized by:

✓ (a) The accumulation of genetic mutations

(b) The breakdown of cellular processes

(c) The cessation of mitosis

(d) The reduction in DNA replication

Answer: (a) The accumulation of genetic mutations — With advancing age, DNA damage and mutations accumulate in cells while repair becomes less efficient. This contributes to the gradual breakdown of cellular processes and the decline in tissue and organ function seen in aging.

6. vi) The inner cell mass of a blastocyst gives rise to:

(a) The placenta

(b) The embryo

(c) The umbilical cord

(d) The amniotic fluid

Answer: (b) The embryo — The blastocyst has an outer trophoblast layer and an inner cell mass. The inner cell mass develops into the embryo proper, while the trophoblast mainly contributes to the placenta and foetal membranes.

7. vii) The process of fertilization occurs in the

(a) Uterus

✓ (b) Fallopian tube

(c) Cervix

(d) Vagina

Answer: (b) Fallopian tube — Fertilization normally takes place in the upper part of the fallopian tube (oviduct), where the sperm meets the egg released at ovulation. The fertilized egg then travels to the uterus for implantation.

8. viii) The stages of embryonic development are

✓ (a) Cleavage, implantation, gastrulation, and organogenesis

(b) Fertilization, implantation, gastrulation, and organogenesis

(c) Cleavage, fertilization, gastrulation, and organogenesis

(d) Cleavage, fertilization, implantation, and organogenesis

Answer: (a) Cleavage, implantation, gastrulation, and organogenesis — After fertilization, the zygote undergoes cleavage to form a blastocyst, which implants in the uterus. Gastrulation then forms the three germ layers, and organogenesis forms the organs from these layers.

✓

9. ix) The placenta is responsible for

(a) Providing oxygen and nutrients to the developing embryo

(b) Removing waste products from the developing embryo

(c) Producing hormones that maintain the pregnancy

✓ (d) All of the above

Answer: (d) All of the above — The placenta transfers oxygen and nutrients from the mother to the embryo and removes carbon dioxide and other waste products from the foetal blood. It also secretes hormones, such as progesterone, that maintain pregnancy, so all the given functions are correct.

10. x) The formation of the neural tube occurs during

(a) Cleavage

(b) Gastrulation

✓ (c) Organogenesis

(d) Implantation

Answer: (c) Organogenesis — During organogenesis the ectoderm folds to form the neural tube, which develops into the brain and spinal cord. This process, called neurulation, is one of the earliest events in organ formation.

## 2. Write short answer of the following:

Question: 11. i) Why cleavage in birds and reptiles called meroblastic?

Answer: In birds and reptiles the egg contains a very large amount of yolk, and cleavage furrows cannot pass through the dense yolk. Only the small disc of cytoplasm on top of the yolk, the blastodisc, divides. Because cleavage is partial and incomplete, involving only part of the egg, it is called meroblastic cleavage.

Question: 12. ii) Why morula is named so?

Answer: The morula is the early embryonic stage formed by cleavage of the zygote, consisting of a solid ball of small cells called blastomeres. Its surface looks like a mulberry fruit (Morula is the Latin name related to mulberry), so it is named morula. It later develops a cavity and changes into the blastula or blastocyst.

Question: 13. iii) How embryonic tissues influence other embryonic tissues?

Answer: Embryonic tissues influence neighbouring tissues by the process of induction, in which one tissue releases chemical signals that direct adjacent cells to differentiate in a particular way. For example, the chordamesoderm induces the overlying ectoderm to form the neural plate and neural tube. Such interactions, controlled by signal molecules and gene switching, coordinate the orderly development of organs.

Question: 14. iv) Define organizers and differentiate between primary and secondary induction.

Answer: An organizer is a group of embryonic cells that influences the development of surrounding tissues and organizes the body plan, such as the dorsal lip of the blastopore in amphibians described by Spemann. Primary induction is the first inductive interaction, in which the chordamesoderm induces the ectoderm to form the nervous system during gastrulation. Secondary induction occurs later between tissues formed after gastrulation, for example the optic vesicle inducing the overlying ectoderm to form the lens of the eye.

Question: 15. v) Differentiate between blastula and gastrula.

Answer: The blastula is an early stage consisting of a hollow ball of cells (blastoderm) around a fluid-filled cavity, the blastocoel or segmentation cavity, and it has no germ layers or gut. The gastrula is the later stage formed by gastrulation, in which cells migrate inward to form three germ layers (ectoderm, mesoderm and endoderm) and a primitive gut, the archenteron. Thus the gastrula marks the beginning of tissue and organ formation.

Question: 16. vi) Why monozygotic twins having same sex?

Answer: Monozygotic twins develop from a single fertilized egg that splits into two embryos early in development. Because both embryos come from the same zygote, they carry identical chromosomes, including the same sex chromosomes (both XX or both XY). They are therefore always of the same sex and are called identical twins.

3. Write detailed answers to the following questions.

Question: 17. i) Explain blastula/blastocyst with emphasis on segmentation cavity.

Answer: After fertilization the zygote undergoes repeated mitotic divisions called cleavage, first forming a solid morula and then the blastula, which in mammals is called the blastocyst. Cleavage creates a fluid-filled space inside, the segmentation cavity or blastocoel, formed as the blastomeres separate and fluid accumulates. In the mammalian blastocyst the cells are arranged as an outer layer, the trophoblast, which will form the placenta and membranes, and an inner cell mass attached at one side, which will form the embryo. The segmentation cavity allows cell movements and rearrangement during the next stage, gastrulation, and in mammals gives the blastocyst the form that implants in the uterine wall.

Question: 18. ii) List the tissues and organs formed from the three germ layers.

Answer: The ectoderm forms the epidermis of the skin with hair, nails and skin glands, the whole nervous system including brain and spinal cord, the sensory epithelia of the eye, ear and nose, the lens and retina, and the enamel of teeth. The mesoderm forms muscles, bones and cartilage, connective tissues, the heart and blood vessels, blood, kidneys and ureters, the reproductive organs, the dermis of skin and the lining of the body cavity. The endoderm forms the lining of the digestive tract and respiratory tract, and the glands derived from them including the liver, pancreas, thyroid and thymus, as well as the lining of the urinary bladder.

Question: 19. iii) Through experimental narration, describe the role of the nucleus and cytoplasm in controlling development.

Answer: Experiments show that both the nucleus and the cytoplasm control development. In nuclear transplantation experiments, for example Briggs and King transplanted a nucleus from an early frog embryo into an enucleated egg and a normal tadpole developed, proving the nucleus carries the complete genetic instructions for development. Gurdon later obtained adult frogs from nuclei of tadpole intestinal cells, showing that differentiated cells retain all genes. On the other hand, the egg cytoplasm contains maternal determinants distributed unevenly, and experiments such as constricting a fertilized egg show that regions lacking certain cytoplasmic substances fail to form particular structures. Development is therefore controlled by the genes of the nucleus acting in response to signals and substances in the surrounding cytoplasm.

Question: 20. iv) Describe the events of development in humans in terms

of first, second and third trimesters.

Answer: In the first trimester (weeks 1-12), fertilization, cleavage, implantation and gastrulation occur, the three germ layers form, and organogenesis builds all major organs; the heart starts beating, limb buds appear and by the end the embryo is called a foetus. In the second trimester (weeks 13-26), the foetus grows rapidly, bones ossify, hair and facial features form, movements are felt by the mother and the sex can be identified, while the organs mature. In the third trimester (weeks 27-40), growth and weight gain are greatest, the lungs mature with surfactant production, the foetus turns head-down, and it becomes capable of independent life at birth.

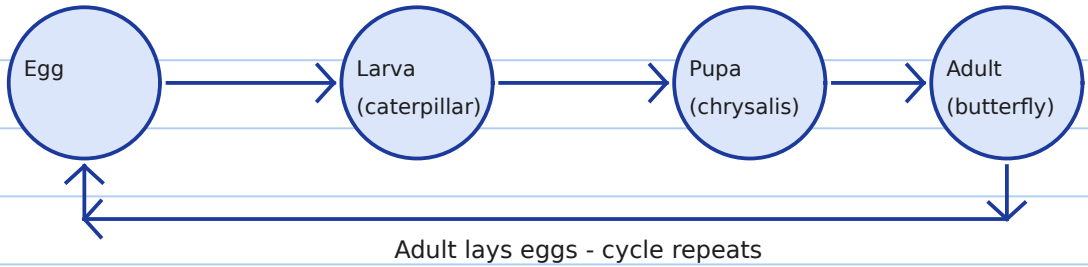



Fig.: Lifecycle (labelled diagram)

Question: 21. v) Describe the maternal derived abnormalities (rubella, abnormal neural tube, thyroid gland and limb development).

Answer: Maternal factors can disturb embryonic development. Rubella (German measles) virus crossing the placenta in early pregnancy can cause deafness, cataracts and heart defects in the baby. Failure of the neural tube to close properly produces abnormalities such as spina bifida and anencephaly, commonly linked with deficiency of folic acid in the mother. Maternal thyroid disorders, especially iodine deficiency and hypothyroidism, impair the baby's brain and skeletal development and can cause cretinism. Drugs and harmful substances taken by the mother, such as thalidomide, can disturb limb development and produce short or missing limbs, showing that a healthy maternal environment is essential for normal development.

# Chapter 22: Chromosome and DNA

## 1. Encircle the correct answer:

1. i) W. Sutton and Theodor observed this behavior of chromosomes.

(a) Formation

✓ (b) Distribution

(c) Staining

(d) Replication

Answer: (b) Distribution — Walter Sutton and Theodor Boveri observed that chromosomes are distributed during meiosis and fertilization in the same pattern as Mendel's hereditary factors. This parallel behaviour led them to propose the chromosomal theory of inheritance.

2. ii) Which is not the part of chromosomal theory of inheritance.

✓ (a) Gametes do not make equal hereditary contribution

(b) Chromosome segregate during meiosis

(c) Nucleus is the room of hereditary material

(d) Gametes have one copy of homologous chromosome

Answer: (a) Gametes do not make equal hereditary contribution — According to the chromosomal theory, both parents make an equal hereditary contribution through their gametes, each gamete carrying one copy of each homologous chromosome. The statement that gametes do not make an equal contribution is therefore not part of the theory.

3. iii) Newly formed chromosome has two.

✓ (a) Chromatids

(b) Heterologous

(c) Centromere

(d) Pairs

Answer: (a) Chromatids — After replication, each chromosome consists of two identical sister chromatids joined at the centromere. These chromatids separate during cell division so that each daughter cell receives one copy.

4. iv) The complex of histone octamer and two loops of duplex DNA is.

(a) Chromomer

(b) Chromatin

✓ (c) Nucleosome

(d) Heterochromatin

Answer: (c) Nucleosome — A nucleosome is the basic unit of chromatin packing, formed by about two loops of duplex DNA wrapped around a core of eight histone proteins (a histone octamer). A chain of nucleosomes gives chromatin its beads-on-a-string appearance.

5. v) Small part of DNA which has information to synthesize specific polypeptide chain is.

(a) Genome

(b) Locus

✓ (c) Gene

(d) Nucleotide

Answer: (c) Gene — A gene is a specific length of DNA that carries the coded information for making one particular polypeptide chain. The genome is the total DNA of the cell, and a locus is simply the position of a gene on a chromosome.

6. vi) According to replication model, the parental double helix remain

intact and conserved called.

(a) Semi conservative

✓ (b) Conservative

(c) Disposed

(d) Eukaryotic replication

Answer: (b) Conservative — In the conservative model, the parental double helix was proposed to remain completely intact and an entirely new double helix to be made. Actual DNA replication was later proved by Meselson and Stahl to be semi-conservative, not conservative.

7. vii) A protein which prevents the re-binding of complementary strand during replication at fork is.

(a) DNA helicase

(b) Ligase

✓ (c) SSB

(d) Primase

Answer: (c) SSB — Single-strand binding proteins (SSB) attach to the separated single strands of DNA at the replication fork. They keep the strands apart, prevent them from re-joining and protect them from damage until new strands are synthesized.

8. viii) The DNA strand which continuously replicate in the direction of 5' to 3' called.

(a) Lagging strand

✓ (b) Leading strand

(c) Primer

(d) Okazaki fragment

Answer: (b) Leading strand — The leading strand is synthesized

continuously in the 5' to 3' direction, in the same direction as the movement of the replication fork. The opposite strand, the lagging strand, is made discontinuously in short Okazaki fragments.

9. ix) Process where information present on gene is used to produce a functional product by living organism called.

(a) Transcription

(b) Translation

✓ (c) Gene expression

(d) Regulation of gene expression

Answer: (c) Gene expression — Gene expression is the overall process by which the information stored in a gene is used to produce a functional product, usually a protein. It includes transcription of the gene into mRNA and translation of mRNA into a polypeptide.

10. x) The set of rules used to store the genetic information within a DNA for particular protein synthesis is.

(a) Gene

✓ (b) Genetic codes

(c) Codon

(d) Anticodon

Answer: (b) Genetic codes — The genetic code is the set of rules by which triplet sequences of bases in DNA and mRNA specify the amino acids of a protein. A codon is a single triplet within this code, while an anticodon is the matching triplet on tRNA.

## 2. Write short answer of the following:

Question: 11. i) Why DNA is negatively charged molecules?

Answer: DNA is negatively charged because each nucleotide of its backbone contains a phosphate group, and phosphate groups carry a negative charge at normal cellular pH. With thousands of phosphate groups along its length, the whole DNA molecule has a strong overall negative charge. This charge helps DNA associate with positively charged histone proteins to form chromatin.

Question: 12. ii) Why replication is called semi-conservative process?

Answer: Replication is called semi-conservative because each daughter DNA molecule formed consists of one old (parental) strand and one newly synthesized strand. In this way half of the original molecule is conserved in each new molecule. The Meselson and Stahl experiment with nitrogen isotopes proved this mode of replication.

Question: 13. iii) Do genes of prokaryotic cell and eukaryotic cells are different in structure? If so give the main differences between them?

Answer: Yes, prokaryotic and eukaryotic genes differ in structure. Prokaryotic genes are continuous, without introns, are often grouped in operons, and lie on a single circular DNA molecule in the nucleoid; transcription and translation occur together in the cytoplasm. Eukaryotic genes are split into coding exons and non-coding introns, each gene is usually transcribed separately, and the DNA is linear and packed with histones in a membrane-bound nucleus, so the primary transcript must be processed before translation.

Question: 14. iv) What are leading and lagging strand of DNA?

Answer: Because DNA polymerase can synthesize DNA only in the 5' to 3' direction and the two template strands are antiparallel, the two new

strands are made differently. The leading strand is synthesized continuously in the direction of fork movement from a single primer. The lagging strand is synthesized discontinuously, away from the fork, as short Okazaki fragments that are later joined by DNA ligase.

Question: 15. v) Give the name of enzymes involved in replication of DNA, also give their brief functions.

Answer: The main enzymes of replication are: helicase, which unwinds and separates the two strands; primase, which lays down a short RNA primer; DNA polymerase III, which builds the new DNA strand; DNA polymerase I, which removes the primer and fills the gap; and ligase, which joins the Okazaki fragments. In addition, SSB proteins keep the single strands separated and topoisomerase relieves the twisting strain ahead of the fork.

Question: 16. vi) Do introns are transcribed, and involve in translation?

Answer: Introns are transcribed, because RNA polymerase copies the whole gene, including both exons and introns, into the primary mRNA transcript. However, introns are not involved in translation, because they are cut out during splicing before the mature mRNA leaves the nucleus. Only the joined exons are read by the ribosome to make the protein.

Question: 17. vii) Give changes occur in mRNA during transport from nucleus to cytoplasm.

Answer: Before leaving the nucleus, eukaryotic mRNA is processed in three ways: a cap is added at the 5' end, a poly-A tail is added at the 3' end, and the introns are spliced out so that the exons join together. This mature mRNA is stable, is protected from degradation, and can be

translated in the cytoplasm. Only after this processing does the mRNA pass through the nuclear pores to the ribosomes.

Question: 18. viii) Give site at ribosome and their functions during translation.

Answer: The large subunit of the ribosome has three sites for tRNA during translation. The A (aminoacyl) site receives the incoming tRNA carrying the next amino acid, the P (peptidyl) site holds the tRNA carrying the growing polypeptide chain, and the E (exit) site is where the empty tRNA leaves the ribosome. The ribosome moves along the mRNA codon by codon, linking amino acids into a chain at these sites.

Question: 19. ix) Draw structure of tRNA and give functions of their different sites.

Answer: A tRNA molecule is folded into a clover-leaf shape with several important regions. At the 3' end is the amino acid attachment site (CCA end), which carries a specific amino acid. At the opposite end is the anticodon loop, whose three-base anticodon pairs with the complementary codon on mRNA. The D loop and T loop help the molecule bind correctly to the ribosome and to its specific activating enzyme, so the tRNA delivers the correct amino acid to the growing chain.

Question: 20. x) Do mutation is always harmful? Justify you answer.

Answer: No, mutation is not always harmful. Many mutations are neutral because they occur in non-coding DNA or do not change the amino acid produced, so they have no visible effect. Some mutations are harmful and cause genetic disorders, but a few are beneficial, giving the organism an advantage in survival and reproduction. Such useful variations are the raw

material of evolution and of plant and animal breeding.

## 3. Give detail answer of following questions:

Question: 21. i) Describe the chromosomal theory of inheritance?

Answer: The chromosomal theory of inheritance, proposed by Sutton and Boveri in 1902, states that genes are located on chromosomes and that chromosomes are the carriers of heredity from one generation to the next. It points out that chromosomes and Mendel's factors both occur in pairs, both segregate during meiosis so that each gamete receives one member of each pair, and both are restored at fertilization when the sperm and egg nuclei fuse. The behaviour of chromosomes in meiosis thus explains Mendel's laws of segregation and independent assortment. The nucleus is the storehouse of this hereditary material, and each parent makes an equal chromosomal contribution to the offspring.

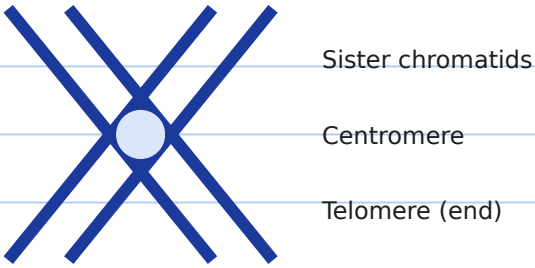
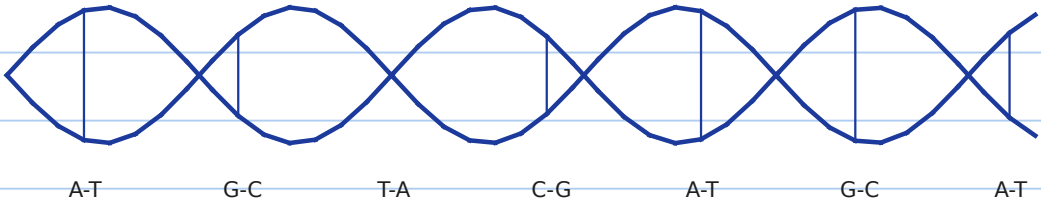



Fig.: Chromosome (labelled diagram)

Question: 22. ii) Describe the semi conservative process of DNA replication?

Answer: In semi-conservative replication the two strands of the parental double helix separate, and each acts as a template for a new complementary strand, so each daughter molecule keeps one old strand and gains one new strand. Helicase unwinds the helix at the origin and a replication fork forms, with SSB proteins holding the strands apart. Primase lays a short RNA primer, DNA polymerase III adds complementary nucleotides continuously on the leading strand and in Okazaki fragments on the lagging strand, and polymerase I and ligase remove primers and seal the joins. Meselson and Stahl proved this mechanism by growing bacteria on heavy nitrogen (N-15) and then on light nitrogen (N-14) and finding DNA of intermediate density after one generation.




Sugar-phosphate backbones with base pairs joined by hydrogen bonds

Fig.: Dna (labelled diagram)

Question: 23. iii) Describe the chemical structure of chromosome.

Answer: A chromosome is made of chromatin, which chemically consists of DNA, histone and non-histone proteins, and a small amount of RNA. About 40 percent is DNA, the hereditary material carrying the genes, and about 60 percent is protein, mainly basic histones around which the DNA is wound to form nucleosomes. At division the chromatin condenses into the short, thick chromosome with two sister chromatids joined at a centromere, with telomeres at the ends. This packing allows the very long DNA molecule to fit inside the nucleus and to be distributed accurately to daughter cells.

Question: 24. iv) Describe the process of transcription during gene expression.

Answer: Transcription is the first step of gene expression, in which the information of one gene on DNA is copied into a complementary messenger RNA by the enzyme RNA polymerase. The polymerase binds at the promoter, unwinds a short length of DNA, and reads the template strand in the 3' to 5' direction, joining complementary RNA nucleotides (with uracil in place of thymine) to build mRNA in the 5' to 3' direction. Transcription continues until a terminator sequence is reached, when the mRNA is released. In eukaryotes the transcript is then capped, tailed and spliced before it leaves the nucleus for translation.

Question: 25. v) What is gene regulation how gene regulate during expression?

Answer: Gene regulation is the turning ON and OFF of genes so that a cell makes only the proteins it needs, at the right time and in the right amount. In prokaryotes it occurs mainly at transcription through operons, where a repressor protein blocks the operator to switch genes off, and removal of the repressor or binding of an activator switches them on, as in the lac operon. In eukaryotes regulation occurs at many levels: chromatin opening, control of transcription by regulatory proteins, mRNA processing, mRNA breakdown and control of translation. Such regulation allows cells with identical genes to differentiate into different types and to respond to changing conditions.

# Chapter 23: Inheritance

## 1. Encircle the correct answer:

1. i) Which of the following would cause phenotypic variations among organisms of the same genotype?

(a) Continuous variation within the species

(b) Different varieties of the same species

(c) Different sexes

✓ (d) Exposure to different environments

Answer: (d) Exposure to different environments - Organisms with the same genotype can look different when they grow under different conditions of food, light, temperature or water. Such variation is environmental and is not inherited.

2. ii) In which of the following examples of human inheritance is the inheritance pattern explained by multiple alleles on an autosomal chromosome?

✓ (a) The ABO blood group system

(b) Cystic fibrosis

(c) Down's syndrome

(d) Haemophilia

Answer: (a) The ABO blood group system - The ABO gene on an autosome has three alleles, IA, IB and i, of which any person carries only two. Different combinations of these alleles produce blood groups A, B, AB and O.

3. iii) What are the phenotypes of the parents of a colour-blind son and a

non-carrier daughter with normal colour vision? Father / Mother:

(a) Carrier / Normal

(b) Colour-blind / Carrier

(c) Colour-blind / Colour-blind

✓ (d) Normal / Carrier

Answer: (d) Normal / Carrier - The son receives his X chromosome from his mother, so the mother must carry the colour-blind allele. The daughter receives her father's X chromosome, so a normal father gives her a normal X and she is a non-carrier when she also receives a normal X from her mother.

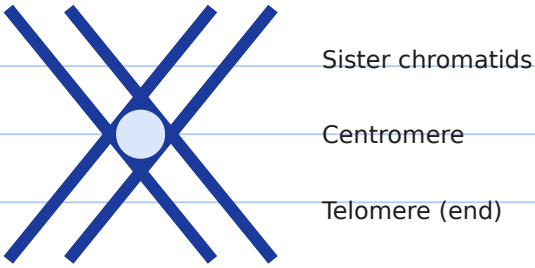



Fig.: Chromosome (labelled diagram)

4. iv) In the F2 generation of a dihybrid cross, the phenotypes occurred in the ratio 3:1 what does this result indicate?

(a) The alleles were segregating independently

(b) Polygenic inheritance was involved

(c) Codominance was being shown

✓ (d) The gene loci were linked

Answer: (d) The gene loci were linked - Linked genes on the same chromosome are inherited together and behave almost like a single gene. That is why a dihybrid cross gives a monohybrid-like 3:1 ratio instead of the usual 9:3:3:1 ratio.

5. v) Two parents, both of blood group A, have a daughter of blood group

O. What is the probability that their next child will be a boy who has blood group O?

(a) 0.25

(b) 0.5

(c) 0.75

(d) 1

Answer: (a) 0.25 - A child with blood group O proves that both group A parents are heterozygous, IAi. In the cross IAi x IAi, one out of four children, that is 0.25, receives i from both parents and has blood group O.

6. vi) A boy is colour blind which could be the genotype of his mother?

(a) XN XN

(b) XN Xn

(c) XN Y

(d) Xn Y

Answer: (b) XN Xn - A boy gets his only X chromosome from his mother, so his mother must carry the recessive colour-blind allele n on one X chromosome. A carrier mother XN Xn is phenotypically normal but passes Xn to half of her sons.

7. vii) Number of chromosome in grass hopper female is 24. How many chromosomes are present in grass hopper male?

(a) 26

(b) 25

(c) 24

(d) 23

Answer: (d) 23 - In grasshopper the female has two X chromosomes (XX) while the male has only one X chromosome (XO) and no second sex

chromosome. The male therefore has one chromosome less, 23 instead of 24.

8. viii) The allele of holandric gene is located at?

(a) X-Chromosome

✓ (b) Y-Chromosome

(c) Autosome

(d) None of these

Answer: (b) Y-Chromosome - Holandric genes are present only on the non-homologous part of the Y chromosome. They pass directly from father to son only, and are always expressed because males are hemizygous for them.

9. ix) The gene which interferes and masks the phenotype of the phenotype of the non-locus gene called?

(a) Mutant gene

✓ (b) Epistatic gene

(c) Pleiotropy

(d) Sex-linked

Answer: (b) Epistatic gene - An epistatic gene present at one locus suppresses or masks the expression of a gene at a different locus. The masked gene is called the hypostatic gene, and this interaction is called epistasis.

10. x) International society of blood transfusion has found more than?

(a) 10 blood system

(b) 20 blood system

✓ (c) 30 blood system

(d) 40 blood system

Answer: (c) 30 blood system - Apart from the well known ABO and Rh systems, the International Society of Blood Transfusion has recognised more than 30 blood group systems in man. Each system is controlled by its own gene or group of genes and its antigens must be considered during transfusion.

## 2. Write short answer of the following:

Question: 11. i) Why Rh-incompatibility could be a danger to the developing foetus and mother?

Answer: An Rh-negative mother carrying an Rh-positive foetus may be sensitised by foetal red cells entering her blood at delivery, and she then makes anti-Rh antibodies. In a later pregnancy these antibodies cross the placenta and destroy the red blood cells of an Rh-positive foetus, causing erythroblastosis foetalis, severe anaemia, jaundice and even death of the foetus. Repeated sensitisation also makes transfusion and later pregnancies risky for the mother.

Question: 12. ii) Why haemophilia is common in human male?

Answer: Haemophilia is an X-linked recessive disorder, and a male has only one X chromosome, so a single recessive allele on that X is expressed and he becomes haemophilic. A female has two X chromosomes, so she is usually only a carrier when one X carries the allele, and she becomes haemophilic only in the rare case when both X chromosomes carry it.

Question: 13. iii) What do you mean by sex-influenced trait?

Answer: A sex-influenced trait is an autosomal trait whose expression differs in males and females because of sex hormones, so the same

genotype behaves as dominant in one sex and recessive in the other. Pattern baldness in man is a common example: it is dominant in males but recessive in females.

Question: 14. iv) Differentiate between following a) Linkage and crossing over b) Monohybrid Cross and Dihybrid Cross c) X-linked trait and Y-linked trait d) Autosomes and sex-chromosome e) Incomplete dominance and Co-dominance

Answer: a) Linkage is the tendency of genes on the same chromosome to be inherited together, while crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes, which produces new combinations. b) A monohybrid cross studies one contrasting trait at a time, while a dihybrid cross studies two traits at the same time. c) An X-linked trait is controlled by a gene on the X chromosome and can appear in both sexes, while a Y-linked trait is controlled by a gene on the Y chromosome and appears only in males. d) Autosomes are chromosomes that are alike in males and females and do not determine sex, while sex chromosomes differ between the sexes and determine sex. e) In incomplete dominance the heterozygote shows a blended, intermediate phenotype, while in co-dominance both alleles are fully and separately expressed in the heterozygote, as in blood group AB.

Question: 15. v) Why human male are heterogametic?

Answer: A human male has two different sex chromosomes, X and Y, so during meiosis he produces two types of sperms: half carrying X and half carrying Y. Because he produces more than one type of gamete for sex, the male is called heterogametic, while the female (XX) produces only X-carrying eggs and is homogametic.

Question: 16. vi) What do you mean by dichromacy and monochromacy?

Answer: Dichromacy is a form of colour blindness in which a person can perceive only two of the three primary colours because one type of cone cells is missing or faulty, for example red blindness (protanopia) or green blindness (deuteranopia). Monochromacy is complete colour blindness, in which the person cannot distinguish colours at all and sees everything in black, white and shades of grey.

Question: 17. vii) What do you known about ZZ and ZW in sex determination?

Answer: In birds, butterflies and some reptiles the sex chromosomes are called Z and W instead of X and Y. The male is homogametic with ZZ chromosomes and produces only Z sperms, while the female is heterogametic with ZW chromosomes and produces Z and W eggs, so in these animals the egg determines the sex of the offspring.

Question: 18. viii) Why incomplete dominance is called partial dominance?

Answer: In incomplete dominance the dominant allele cannot completely suppress the recessive allele in the heterozygote, so the offspring shows a phenotype midway between the two parents, such as pink flowers from red and white parents in Mirabilis. Because dominance is only partial and not complete, it is also called partial dominance.

Question: 19. ix) Define following terms (a) Genetics (b) Allele (c) Gene (d) Karyotype (e) multiple allele (f) test cross (g) Mutant (h) Test cross (i) Back cross (j) Linkage (k) Crossing over (l) Protanopi (m) Isoagglutinogen (n) Erythroblastosis foetilus

Answer: (a) Genetics is the branch of biology that studies heredity and

variation. (b) An allele is an alternative form of a gene present at the same locus on homologous chromosomes. (c) A gene is a specific part of DNA that controls one hereditary character. (d) Karyotype is the complete set of chromosomes of a cell shown by number, size and shape. (e) Multiple alleles are three or more alternative forms of the same gene present in a population, such as IA, IB and i for blood groups. (f) and (h) A test cross is a cross of an individual of unknown genotype showing the dominant phenotype with a homozygous recessive individual, to find out its genotype. (g) A mutant is an organism or gene that has undergone a sudden heritable change called mutation. (i) A back cross is a cross of the F1 hybrid with either of its parents. (j) Linkage is the tendency of genes located on the same chromosome to be inherited together. (k) Crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes during meiosis. (l) Protanopia (printed Protanopi) is red colour blindness, in which a person cannot perceive red colour. (m) Isoagglutinogens are the antigens (A and B) present on the surface of red blood cells that cause agglutination with the opposite antibody. (n) Erythroblastosis foetalis is the disease of a newborn Rh-positive baby of an Rh-negative mother, in which maternal anti-Rh antibodies destroy the baby's red blood cells.

Question: 20. x) Why law of segregation is called purity of gametes?

Answer: According to the law of segregation, the two alleles of a gene separate during gamete formation, so each gamete receives only one allele and remains pure for that character. A gamete never carries both alleles together, therefore this law is called the law of purity of gametes.

## 3. Give detail answer of following questions:

Question: 21. i) Explain epistasis with reference to the inheritance of coat colour in Labrador retriever.

Answer: Epistasis is the interaction in which a gene at one locus masks the expression of a gene at another locus. In Labrador retrievers, gene B controls the colour of the pigment: BB or Bb gives black coat and bb gives chocolate (brown) coat. Another gene, E, controls whether the pigment is deposited in the fur at all: E allows deposition, but ee prevents it and the dog is yellow, whatever the B genotype. Thus a dog with B_ee or bbee is yellow, B_E_ is black and bbE_ is chocolate, giving a 9:3:4 ratio in F2 instead of 9:3:3:1. Here gene E is epistatic over gene B.

Question: 22. ii) Describe inheritance of two traits with the help of genetic cross.

Answer: Inheritance of two traits is studied by a dihybrid cross, for example Mendel's cross between pea plants with round yellow seeds (RRYY) and wrinkled green seeds (rryy). All F1 plants are heterozygous (RrYy) with round yellow seeds, showing that round and yellow are dominant. On self-crossing, F1 gives F2 in the ratio 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green, because each gene pair segregates and assorts independently, producing four types of gametes (RY, Ry, rY, ry) from each parent. The appearance of new combinations, round green and wrinkled yellow, proves the law of independent assortment.

Question: 23. iii) Describe ABO blood system in human with genetic cross.

Answer: The ABO system is controlled by a single gene with three alleles: IA and IB are co-dominant to each other and both are dominant over i. The genotypes IAIA or IAi give group A, IBIB or IBi give group B, IAIB gives

group AB (both antigens A and B are present), and ii gives group O (no antigen). For example, a cross between a group A father (IAi) and a group B mother (IBi) can produce children of all four groups: IAIB (AB), IAi (A), IBi (B) and ii (O), each with a 25 percent chance. This shows that two parents neither of whom is AB or O can still have AB and O children.

Question: 24. iv) Explain sex determination in human and sex linked (x-linked) trait with genetic cross.

Answer: In humans females are XX and males are XY. The mother always passes an X chromosome to the child, while the father passes either X, producing a daughter (XX), or Y, producing a son (XY), so the father's sperm determines the sex of the baby in roughly a 50:50 ratio. Genes present on the X chromosome show X-linked inheritance, for example colour blindness: a carrier mother (XN Xn) crossed with a normal father (XN Y) gives normal daughters, half of them carriers, normal sons, and colour-blind sons (Xn Y) in equal proportions. Sons therefore get X-linked recessive disorders from their carrier mothers, while daughters are affected only when they receive the allele from both parents.

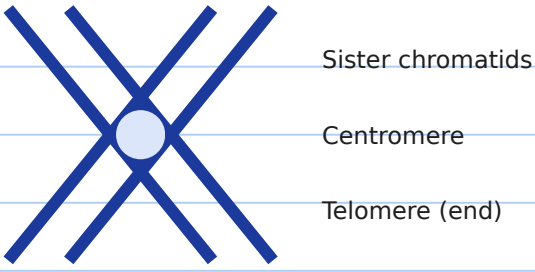



Fig.: Chromosome (labelled diagram)

Question: 25. v) A male has co-dominant blood group marry with female whose blood group is homozygous A, both have a male child who marry with a female with recessive blood group what will be the chances of blood group in their children? Prove it with genetic cross.

Answer: The co-dominant blood group of the male is AB (IAIB) and the female is homozygous A (IAIA). Their

children receive IA from the mother and either IA or IB from the father, so all their children are either IAIA (group A) or IAIB (group AB), each with 50 percent chance; their male child is therefore either group A (IAIA) or group AB (IAIB). When this son marries a female with the recessive blood group O (ii), she gives i to every child: if the son is IAIA, all children are IAi, that is 100 percent group A; if the son is IAIB, half the children are IAi (group A) and half are IBi (group B). So the grandchildren can only be group A or group B, and group O or AB is not possible in this cross.

# Chapter 24: Evolution

## 1. Encircle the most appropriate response.

1. i) Which of the following was supporting the theory of special creation?

(a) George Buffon

(b) Lamarck

(c) Darwin

✓ (d) Suarez

Answer: (d) Suarez - Suarez was a supporter of the theory of special creation, according to which all species were created in their present form by a divine power. Buffon, Lamarck and Darwin, in contrast, presented ideas that supported change in species over time.

2. ii) Archaeopteryx is a connecting link between

(a) Amphibia and reptiles

✓ (b) Reptiles and Aves

(c) Aves and mammals

(d) Fish and amphibia

Answer: (b) Reptiles and Aves - Archaeopteryx had reptilian features such as teeth, claws and a long bony tail, along with bird features such as feathers and wings. It therefore links reptiles with Aves (birds) and shows that birds evolved from reptiles.

3. iii) All of the following are homologous organs except

(a) Wings of bat

✓ (b) Wings of butterfly

(c) Wings of bird

(d) Flippers of dolphin

Answer: (b) Wings of butterfly - The wings of bat and bird and the flippers of dolphin are all modified vertebrate forelimbs with the same basic bone pattern, so they are homologous. The wing of a butterfly is made of chitin and has no such bones, so it is not homologous with them.

4. iv) Which of the following are analogous organs except:

(a) Wings of insect

(b) Wings of bird

(c) Wings of bat

✓ (d) Arm of man

Answer: (d) Arm of man - The wings of an insect, a bird and a bat perform the same function of flight but have different structure and origin, so they are analogous to each other. The arm of man is not a flight organ and is homologous, not analogous, to the vertebrate wings.

5. v) Life originated in

(a) Air

✓ (b) Water

(c) Land

(d) Space

Answer: (b) Water - The first life forms originated in water, especially in shallow seas and hot springs called hydrothermal vents. Water provided the medium in which simple chemicals combined to form complex organic compounds and the first cells.

6. vi) Philosophie Zoologique was written by

(a) Wallace

(b) Darwin

(c) Weismann

✓ (d) Lamarck

Answer: (d) Lamarck - Jean Baptiste de Lamarck published his theory of evolution in his book Philosophie Zoologique in 1809. In it he proposed the inheritance of acquired characters through use and disuse of organs.

7. vii) Which of the following is incorrect regarding Lamarck's theory of Inheritance of acquired characters?

(a) Use and disuse of the organs inherited

(b) Continuously using some organ results in its further strengthening in offspring

(c) Continuously disuse of some organ results in its weakening in offspring

✓ (d) Concrete support to the theory through experiments

Answer: (d) Concrete support to the theory through experiments - Lamarck's ideas about use, disuse and strengthening or weakening of organs in the offspring were his proposed explanation, but they were never supported by concrete experimental evidence. Later experiments, such as those of Weismann, showed that characters acquired during life are not inherited.

8. viii) The theory on human population which inspired Darwin was proposed by

✓ (a) Malthus

(b) Lyell

(c) Hutton

(d) Wallace

Answer: (a) Malthus - Thomas Malthus proposed that human population increases faster than its food supply, so many individuals must die in the struggle for existence. Darwin applied this idea to all species and developed the idea of survival of the fittest through natural selection.

9. ix) Theory of Natural Selection was lacking any support from

(a) Biogeography

✓ (b) Genetics

(c) Comparative anatomy

(d) Molecular Biology

Answer: (b) Genetics - Darwin supported natural selection with evidence from biogeography, comparative anatomy and fossils, but he did not know the mechanism of heredity. Genetics was not developed in his time, so his theory lacked genetic support until Neo-Darwinism combined it with genetics.

10. x) Which of the following cannot be the disruptive force for Hardy-Weinberg Theorem?

(a) Selection

(b) Non-random mating

(c) Mutation

✓ (d) Random mating

Answer: (d) Random mating - The Hardy-Weinberg equilibrium stays undisturbed when mating is random, there is no selection, no mutation, no migration and the population is large. Selection, non-random mating and mutation change allele frequencies, so they disrupt the equilibrium, while random mating does not.

## 2. Write short answers of the following questions.

Question: 11. i) Differentiate between the following: a) Special Creation and Evolution b) Lamarckism and Darwinism c) Invagination theory and Endosymbiotic theory d) Bottle-neck effect and Founder's effect e) Convergent and Divergent evolution

Answer: a) Special creation holds that species were created in their present form and do not change, while evolution holds that species change gradually over time and new species arise from earlier ones. b) Lamarckism explains evolution by use and disuse of organs and inheritance of acquired characters, while Darwinism explains it by variation and natural selection of the fittest. c) The invagination theory says eukaryotic organelles formed when the prokaryotic cell membrane folded inwards to enclose the genetic material, while the endosymbiotic theory says mitochondria and chloroplasts arose from free-living bacteria that were taken inside another cell and lived in symbiosis. d) In the bottleneck effect a disaster greatly reduces the size of a population and its genetic variation, while in the founder effect a small group separates from a large population and starts a new population with a different gene pool. e) In convergent evolution unrelated organisms develop similar features because of similar environments, while in divergent evolution related organisms develop different features from a common ancestor.

Question: 12. ii) How the homologous organs support the theory of evolution?

Answer: Homologous organs have the same basic structure and embryonic origin but perform different functions, such as the forelimbs of man, bat, bird and whale. Such similarity of plan in different groups can only be explained if they inherited the structure from a common ancestor and then

modified it for different uses. This strongly supports evolution and divergent change from common ancestry.

Question: 13. iii) What are analogous organs? Give example.

Answer: Analogous organs are organs that perform the same function but have different structure, origin and ancestry, such as the wings of an insect and the wings of a bird. Both are used for flying, but the insect wing is a chitinous outgrowth without bones, while the bird wing is a modified forelimb with bones. Their similarity is due to convergent evolution under similar conditions, not common ancestry.

Question: 14. iv) How Alfred Wallace contributed Charles Darwin regarding the Natural Selection?

Answer: Alfred Russel Wallace, working independently in the Malay Archipelago, reached the same idea of natural selection as Darwin and sent his essay to Darwin in 1858. This contribution pushed Darwin to finally publish his work, and their papers were presented together, after which Darwin published On the Origin of Species in 1859. Wallace also contributed greatly through his studies of the geographical distribution of animals.

Question: 15. v) What do understand by the "descent with modification"?

Answer: Descent with modification is Darwin's phrase for evolution: present-day species have descended, with gradual changes, from pre-existing ancestral species. Over long periods these small inherited modifications accumulate, producing organisms that are different from their ancestors and, in time, entirely new species.

Question: 16. vi) How Neo-Darwinism differs from Darwinism?

Answer: Darwinism explains evolution mainly by natural selection acting on variations, but Darwin could not explain the origin and inheritance of variations. Neo-Darwinism is the modern form of the theory that adds genetics: variations arise by mutation and recombination, are stored in the gene pool of a population, and natural selection, genetic drift and isolation change allele frequencies to produce new species.

Question: 17. vii) How the sympatric speciation differs from parapatric speciation?

Answer: In sympatric speciation a new species arises within the same geographical area, without any physical barrier, often through polyploidy or a change in habits or mating time. In parapatric speciation a new species arises in a population that lives in neighbouring, partly overlapping areas, where gene flow is limited and the two ends of the range gradually become different species.

Question: 18. viii) Justify Lamarck as an early proponent of evolution.

Answer: Lamarck was one of the first scientists to propose, in 1809, a complete theory that species are not fixed but change over time to become better adapted to their environment. Although his explanation of inheritance of acquired characters was later rejected, he introduced the ideas of gradual change, adaptation and use and disuse, and prepared the way for Darwin. He is therefore rightly regarded as an early proponent of evolution.

Question: 19. ix) What is genetic drift?

Answer: Genetic drift is the random change in allele frequencies in a

population from one generation to the next, occurring purely by chance. It is strongest in small populations, where a few individuals breeding or dying by chance can greatly change or even eliminate an allele, as seen in the bottleneck and founder effects.

Question: 20. x) List out the factors affecting allele frequency

Answer: The main factors that change allele frequency in a population are mutation, natural selection, genetic drift, gene flow (migration of individuals in or out of the population) and non-random mating. Any of these factors disturbs the Hardy-Weinberg equilibrium and can lead to evolution of the population over generations.

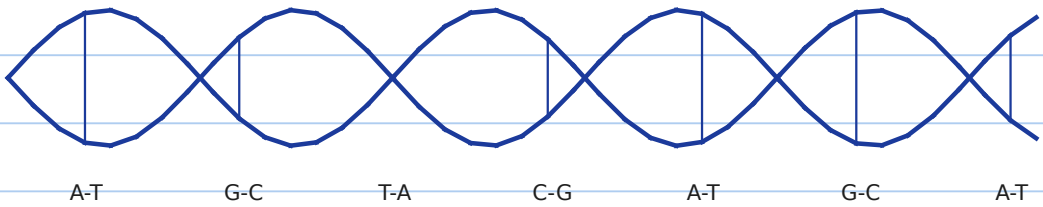
## 3. Write detailed answers to the following questions.

Question: 21. i) State and explain the contribution of Lamarck in organic evolution.

Answer: Jean Baptiste de Lamarck presented the first systematic theory of organic evolution in his book Philosophie Zoologique (1809), known as Lamarckism or the theory of inheritance of acquired characters. He proposed that a changing environment creates new needs in organisms, that organs which are used more become stronger and larger while disused organs weaken and may disappear, and that these characters acquired during life are inherited by the offspring, his famous example being the long neck of the giraffe developed by stretching to reach high leaves. His mechanism was later disproved, but his contribution remains important because he first established that species change gradually, that adaptation is related to the environment, and that evolution proceeds from simple to complex forms.

Question: 22. ii) Discuss the process of origin of single cell Eukaryotes.

Answer: Single-celled eukaryotes arose from prokaryotic ancestors more than about 1.5 billion years ago in water. According to the invagination theory, the cell membrane of a prokaryote folded inwards and enclosed the genetic material, forming a true nucleus and internal membranes such as the endoplasmic reticulum. According to the endosymbiotic theory, larger prokaryotic cells engulfed smaller aerobic bacteria, which were not digested but lived inside them in symbiosis and became mitochondria, and photosynthetic cyanobacteria similarly became chloroplasts. These changes gave the first single-celled eukaryotes a nucleus, organelles and greater efficiency, from which all later eukaryotic life evolved.




Sugar-phosphate backbones with base pairs joined by hydrogen bonds

Fig.: Dna (labelled diagram)

Question: 23. iii) How biogeography and paleontology provide evidences in support of evolution?

Answer: Biogeography, the study of the distribution of species, shows that organisms on islands and continents resemble the nearest mainland forms yet differ in a gradual way, such as the finches of the Galapagos Islands, and that similar environments on different continents have very different species; this fits descent with modification after continents separated by plate tectonics, not independent creation everywhere. Paleontology, the study of fossils, provides direct historical evidence: fossils in older rocks are simpler than those in younger rocks, extinct forms such as dinosaurs are found, and connecting links such as Archaeopteryx and series like the evolution of the horse show step-by-step change from ancestors to modern forms. Together, both lines of evidence strongly support evolution.

Question: 24. iv) What is speciation? Explain different ways of speciation.

Answer: Speciation is the biological process by which one or more new species are formed from an existing species when populations become reproductively isolated and accumulate genetic differences. In allopatric speciation, a geographical barrier such as a river, mountain or sea separates a population, and the separated parts evolve independently into new species. In sympatric speciation, a new species arises within the same area without a barrier, for example by polyploidy in plants or a change in food, habitat or breeding season. In parapatric speciation, neighbouring populations in adjoining areas with limited gene flow gradually diverge into separate species.

Question: 25. v) Describe the Theory of Natural Selection.

Answer: Darwin's theory of natural selection is based on a few observations: organisms produce far more offspring than can survive, resources are limited so there is a struggle for existence, individuals of a population show heritable variations, and organisms are not all equally fit for their environment. Individuals with variations favourable for that environment survive and reproduce more successfully - survival of the fittest - and pass their useful characters to their offspring, while the less fit are gradually eliminated. Continued over many generations, this natural selection accumulates favourable variations and produces new, better adapted species, which is the process Darwin called descent with modification.

# Chapter 25: Man and his Environment

## 1. Encircle the correct choice.

1. i) Which one of these mainly causes the greenhouse effect?

(a) Acid rain

(b) Ozone layer depletion

✓ (c) Global warming

(d) Forest fire

Answer: (c) Global warming - The greenhouse effect is the trapping of heat in the atmosphere by greenhouse gases such as carbon dioxide, and its increase is seen as global warming. As the Earth becomes warmer, ice melts, sea level rises and weather patterns change.

2. ii) Factors that affect the flow of energy at the trophic level are

(a) Non-predatory deaths

(b) Growth & Reproduction

✓ (c) Heat loss

(d) Sunlight

Answer: (c) Heat loss - At every trophic level a large part of the energy is lost as heat during respiration and other life activities. Because of this heat loss, only about 10 percent of the energy of one trophic level passes to the next level.

3. iii) The main cause of the increase in the amount of $CO_2$ in the Earth's atmosphere is

(a) Rapidly growing human population

✓ (b) Burning of a large number of fossil fuels

(c) Increase in the number of industries

(d) Decrease in the forested resources.

Answer: (b) Burning of a large number of fossil fuels - Burning of coal, oil and gas in vehicles, factories and power houses releases huge amounts of carbon dioxide into the air. This is the main cause of the steady rise of $CO_2$ in the atmosphere and of the greenhouse effect.

4. iv) All is positive for which of the following processes?

(a) Oxidation of Nitrogen

✓ (b) Melting of ice and evaporation of water

(c) Oxidation of gold

(d) Burning of chlorine

Answer: (b) Melting of ice and evaporation of water - Melting of ice and evaporation of water absorb heat energy from the surroundings, so the energy change for these processes is positive (endothermic). Such processes take in energy instead of releasing it.

5. v) Which of the following dietary requirements is involved in nitrogen balance?

(a) Carbohydrates

(b) Vitamins

✓ (c) Proteins

(d) Essential fatty acids

Answer: (c) Proteins - Nitrogen enters the body mainly through proteins in food and is lost mainly in urea in urine. Nitrogen balance compares this intake and loss of nitrogen, so it depends directly on the amount of protein eaten.

6. vi) Which of the following cycle is not a gaseous type of cycle?

(a) Carbon cycle

(b) Nitrogen cycle

✓ (c) Phosphorus cycle

(d) Oxygen cycle

Answer: (c) Phosphorus cycle - In the phosphorus cycle the main reservoir is rocks and soil, not the atmosphere, so it is a sedimentary cycle. Carbon, nitrogen and oxygen cycles are gaseous cycles because their main reservoir is the atmosphere or oceans.

7. vii) The source of carbon to plants in the carbon cycle is

(a) Carbonate rocks

✓ (b) atmospheric carbon dioxide

(c) Fossil fuels

(d) all of the above

Answer: (b) atmospheric carbon dioxide - Green plants take carbon dioxide directly from the atmosphere for photosynthesis and convert it into food. From plants this carbon passes to animals and other organisms through food chains.

8. viii) Conversion of nitrates to nitrogen is called

(a) Nitrification

✓ (b) Denitrification

(c) Ammonification

(d) Nitrogen fixation

Answer: (b) Denitrification - In denitrification, denitrifying bacteria convert nitrates of the soil back into free gaseous nitrogen, which returns to the atmosphere. It is the reverse of nitrification, in which ammonia is

converted into nitrites and nitrates.

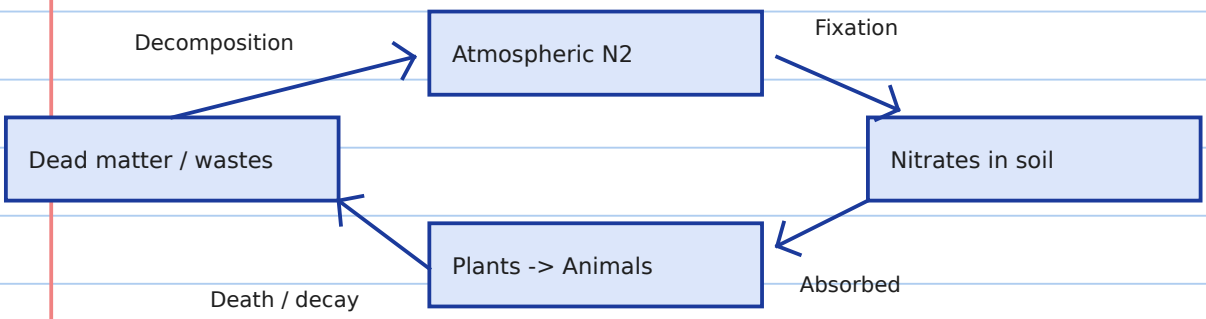



Fig.: Nitrogen (labelled diagram)

9. ix) what is the only source of energy for all ecosystems on Earth?

(a) Water

✓ (b) Sun

(c) Animals

(d) Plants

Answer: (b) Sun - Solar energy is the ultimate and only original source of energy for almost all ecosystems on Earth. Producers capture it by photosynthesis, and from them energy flows to all consumers through food chains.

10. x) The pioneers in xerarch succession are the

(a) Foliose lichens

(b) Mosses

✓ (c) Crustose lichens

(d) Shrubs

Answer: (c) Crustose lichens - Xerarch succession starts on dry bare rock, where crust-like lichens are the first, pioneer organisms to grow. They slowly break the rock surface and help in soil formation, preparing the way for mosses and higher plants.

## 2. Write short answers of the following questions.

Question: 11. i) Why biogeochemical cycle is named so?

Answer: It is called biogeochemical because the elements involved move in a cycle through living organisms (bio), the Earth and soil (geo) and chemical changes in air and water (chemical). For example, nitrogen passes from air to soil, into plants and animals, and back to air through the activities of living organisms and chemical processes.

Question: 12. ii) Differentiate between nitrification and de-nitrification?

Answer: Nitrification is the conversion of ammonia into nitrites and then nitrates by nitrifying bacteria such as Nitrosomonas and Nitrobacter, and it makes nitrogen available to plants in the soil. Denitrification is the opposite process, in which denitrifying bacteria convert nitrates back into free nitrogen gas, which escapes into the atmosphere and reduces soil fertility.

Question: 13. iii) How does energy flow between trophic levels?

Answer: Energy flows in one direction only, from producers to herbivores, then to carnivores and finally to decomposers, through food chains and food webs. At each transfer only about 10 percent of the energy is passed on, while the rest is lost as heat, used in life activities or left in undigested material, so higher trophic levels receive much less energy.

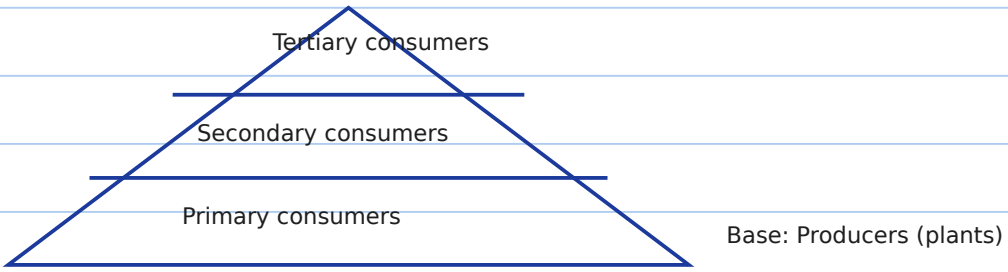



Fig.: Foodchain (labelled diagram)

Question: 14. iv) What are the major producers in a terrestrial ecosystem?

Answer: The major producers in a terrestrial ecosystem are green plants, including grasses, herbs, shrubs and trees, which make food by photosynthesis. In a forest the trees are the chief producers, while in grassland the grasses play this role. All other organisms of the ecosystem depend directly or indirectly on them for food and energy.

Question: 15. v) How does ecological succession affect the community?

Answer: Ecological succession gradually changes the community of an area: simple pioneer species are replaced step by step by more complex and larger species through seral stages. It increases the diversity, biomass and stability of the community and finally produces a climax community that is in balance with the climate of the area.

Question: 16. vi) What are the causes of ozone layer depletion?

Answer: The main cause of ozone layer depletion is the release of chlorofluorocarbons (CFCs) used in refrigerators, air conditioners, sprays and foams. Other ozone-destroying substances include halons, carbon tetrachloride and nitrogen oxides from jet planes and fertilisers. In the upper atmosphere these release chlorine, which breaks down ozone molecules and thins the protective ozone layer.

Question: 17. vii) Why should we conserve biodiversity?

Answer: Biodiversity provides us with food, medicines, fibres and many other useful products and maintains ecosystem services such as pollination, soil formation, water purification and balance of gases in the air. Loss of species can collapse food webs and make ecosystems unstable, so conserving biodiversity is essential for human survival and well-being.

Question: 18. viii) Differentiate between renewable and non-renewable resources.

Answer: Renewable resources are replaced naturally in a short time and can be used again and again if used carefully, such as sunlight, water, air, forests and animals. Non-renewable resources are present in fixed amounts and are not replaced after use within a human lifetime, such as coal, oil, natural gas and minerals, so they must be used wisely and conserved.

## 3. Write the details answers of the following questions:

Question: 19. i) What is a biogeochemical cycle? Explain nitrogen cycle in detail.

Answer: A biogeochemical cycle is the repeated circulation of an element, such as carbon or nitrogen, between living organisms and the non-living environment, so that the element is used again and again. In the nitrogen cycle, atmospheric nitrogen (about 78 percent of air) is first fixed into compounds by nitrogen-fixing bacteria in soil and in the root nodules of legumes, by lightning and by industry, forming ammonia and nitrates. Plants absorb nitrates and make proteins, animals obtain nitrogen by eating plants, and the dead bodies and wastes of organisms are broken down by decomposers into ammonia, a step called ammonification. Nitrifying bacteria convert ammonia into nitrites and nitrates (nitrification), which plants absorb again, while denitrifying bacteria convert some nitrates back into free nitrogen gas (denitrification), completing the cycle.

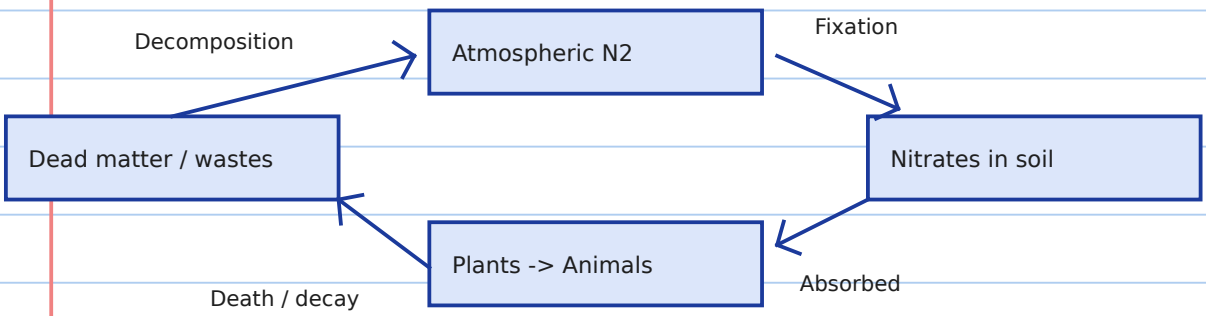



Fig.: Nitrogen (labelled diagram)

Question: 20. ii) Describe the effects of Carbon dioxide and global warming on human health.

Answer: Increase of carbon dioxide causes the greenhouse effect and global warming, which affect human health in many ways. Higher temperatures cause heat strokes and dehydration, especially in old people and outdoor workers, while warmer and wetter conditions help mosquitoes and other vectors spread, increasing malaria, dengue and other infectious diseases. Global warming also disturbs rainfall and agriculture, causing food shortages and malnutrition, and melting ice and rising sea level displace coastal populations. Increased air pollution and allergens along with warming worsen asthma and other breathing problems.

Question: 21. iii) What is succession? Describe different stages of secondary succession with examples.

Answer: Succession is the gradual and directional process by which the community of an area changes over time until a stable climax community is formed. Secondary succession occurs where a community has been disturbed but soil and some organisms remain, for example a burnt forest or an abandoned cultivated field. Its stages are: first annual weeds and grasses grow, then perennial herbs and shrubs appear, followed by fast-growing softwood trees such as pine, and finally hardwood trees form the climax forest. Because soil is already present, secondary succession is much faster than primary succession.

Question: 22. iv) What factors determine the growth rate of the human population? Give their reason also

Answer: The growth rate of the human population is determined mainly by birth rate and death rate, and also by immigration and emigration. A high birth rate increases population because more children are added, while better medical facilities, sanitation and food supply have reduced the death rate, so people live longer and population grows faster. Immigration increases the population of the area people move into, while emigration decreases it. Improvements in agriculture, industry and control of diseases are the reasons why death rate has fallen much faster than birth rate, causing rapid population growth.

# Chapter 26: Biotechnology

## 1. Encircle the correct choice.

1. i) Cloning of genes creates an identical copy of a DNA sequence. The process of cloning is significant for what reason?

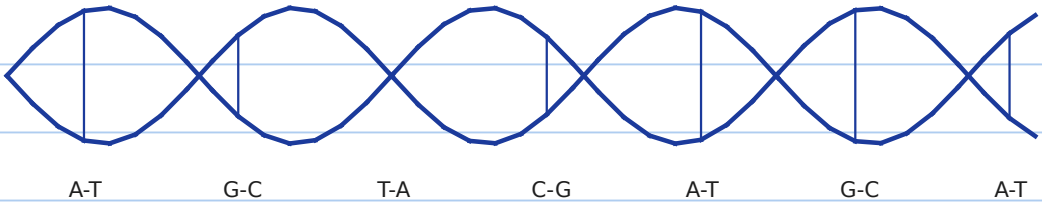
(a) To create a transgenic organism

(b) To diagnose genetic disorders

(c) To study gene function

✓ (d) All the above

Answer: (d) All the above - Cloned genes provide many identical copies of a gene, which can be inserted into organisms to create transgenic organisms and used as probes to diagnose genetic disorders. Cloning also allows scientists to study the function of a gene in detail.




Sugar-phosphate backbones with base pairs joined by hydrogen bonds

Fig.: Dna (labelled diagram)

2. ii) DNA sequencing is a technique used to determine the sequence of nucleotides in a DNA molecule. What is the significance of DNA sequencing?

✓ (a) To identify genetic variations

(b) To create a transgenic organism

(c) To study gene expression

(d) All the above

Answer: (a) To identify genetic variations - By reading the exact order of nucleotides, sequencing reveals changes or mutations in DNA when a sequence is compared with a normal reference sequence. These identified

variations help in diagnosing genetic diseases and in studying differences between individuals and species.

3. iii) DNA analysis is used to identify genetic disorders. What is the purpose of DNA analysis?

(a) Sequence of nucleotides in a DNA

(b) To study gene expression

✓ (c) To identify genetic variations

(d) All the above

Answer: (c) To identify genetic variations - DNA analysis examines a sample of DNA to detect variations, mutations or specific markers present in it. Finding such variations makes it possible to identify genetic disorders, carriers of a disease and relationships between individuals.

4. iv) Biotechnology has many applications in healthcare, including gene therapy, drug production, and diagnosis of diseases. What is the significance of biotechnology in healthcare?

✓ (a) To study gene function

(b) To improve animal welfare

(c) To develop new materials

(d) All the above

Answer: (a) To study gene function - Studying how genes work and how faulty genes cause disease is the basis of the healthcare uses of biotechnology. This knowledge makes gene therapy, production of drugs such as insulin, and accurate molecular diagnosis of diseases possible.

5. v) Polymerase chain reaction (PCR) is a technique used to amplify DNA. What is the significance of PCR?

(a) To sequence DNA

(b) To identify genetic disorders

(c) To study gene expression

✓ (d) All the above

Answer: (d) All the above - PCR makes millions of copies of a small DNA sample, providing enough DNA for sequencing. The amplified DNA is also used to identify genetic disorders and, in its reverse-transcriptase form, to study gene expression.

6. vi) Gel electrophoresis is a technique used to separate DNA fragments based on size. What is the significance of gel electrophoresis?

(a) To study gene expression

(b) To sequence DNA

✓ (c) To identify genetic variations

(d) All the above

Answer: (c) To identify genetic variations - DNA fragments move through the gel at different speeds according to their size, forming bands that can be compared between samples. Differences in band patterns reveal genetic variations, which is used in DNA fingerprinting and diagnosis.

7. vii) Sanger sequencing is a technique used to read the sequence of DNA. What is the significance of Sanger sequencing?

(a) To identify genetic disorders

✓ (b) To sequence DNA

(c) To study gene expression

(d) All the above

Answer: (b) To sequence DNA - Sanger sequencing determines the exact order of nucleotides in a DNA fragment by using chain-terminating

nucleotides. It was the standard method used to sequence genes and even whole genomes, including in the Human Genome Project.

8. viii) Southern blotting is a technique used to transfer DNA fragments from a gel to a membrane. What is the significance of Southern blotting?

✓ (a) To identify genetic disorders

(b) To sequence DNA

(c) To study gene expression

(d) All the above

Answer: (a) To identify genetic disorders - In Southern blotting, DNA fragments fixed on a membrane are detected with a labelled probe that binds to a specific sequence. This detects mutations and abnormal gene fragments, so it is used to identify genetic disorders.

9. ix) Genomic maps are useful for:

✓ (a) Identifying genes associated with specific diseases

(b) Creating genetically modified organisms

(c) Analyzing DNA methylation patterns

(d) Detecting protein-protein interactions

Answer: (a) Identifying genes associated with specific diseases - A genomic map shows the location of genes and markers on chromosomes. By comparing maps of healthy and diseased persons, scientists can locate and identify the genes responsible for specific diseases.

10. x) What is the significance of monoclonal antibodies in biotechnology?

(a) To study gene expression

✓ (b) Used to diagnose diseases

(c) Genetically modified organisms

(d) They can be used to create new drugs

Answer: (b) Used to diagnose diseases - Monoclonal antibodies are identical antibodies that bind to one specific antigen, so they detect that substance very accurately. They are widely used in diagnostic tests, for example pregnancy tests and tests for infections and cancers.

## 2. Write short answer of the following:

Question: 11. i) Why AmpR and LacZ are used in construction of rDNA?

Answer: AmpR and LacZ are used as selectable and screenable marker genes in the plasmid vector during construction of recombinant DNA. AmpR gives resistance to the antibiotic ampicillin, so only bacteria that have taken up the plasmid can grow on ampicillin medium. The LacZ gene produces an enzyme that gives blue colonies; when foreign DNA is inserted into LacZ the gene is inactivated and the colonies stay white, so recombinant bacteria can be recognised by blue-white screening.

Question: 12. ii) What do you mean by RFLPs?

Answer: RFLPs (restriction fragment length polymorphisms) are differences in the lengths of DNA fragments produced when DNA of different individuals is cut with the same restriction enzyme. These differences arise because mutations change the cutting sites of the enzyme from person to person. RFLP patterns are used as genetic markers in DNA fingerprinting, disease diagnosis and mapping of genes.

Question: 13. iii) How is biotechnology used in healthcare, and what are some of the potential benefits and challenges associated with its use?

Answer: Biotechnology is used in healthcare for producing medicines such as insulin, growth hormone and clotting factors in transgenic bacteria, for

gene therapy, for vaccines, monoclonal antibodies and for molecular diagnosis of diseases before symptoms appear. Its benefits are safer and cheaper medicines, accurate early diagnosis and treatment of previously incurable genetic diseases. Its challenges include high cost, ethical questions about altering genes, possible side effects and the risk of misuse of genetic information.

Question: 14. iv) Why restriction enzymes are called molecular scissors?

Answer: Restriction enzymes cut DNA at specific palindromic sequences, just as scissors cut at a chosen point. Each enzyme recognises its own specific sequence and cuts both strands there, often producing sticky ends. Because they cut DNA molecules precisely at desired sites for joining with other DNA, they are called molecular scissors.

Question: 15. v) What do you mean by palindrome?

Answer: A palindrome in DNA is a sequence of bases that reads the same on both strands when read in the 5 to 3 direction, like the word MADAM reads the same from both sides. For example, the sequence GAATTC on one strand pairs with CTTAAG on the other and reads GAATTC in both directions. Restriction enzymes recognise and cut such palindromic sequences.

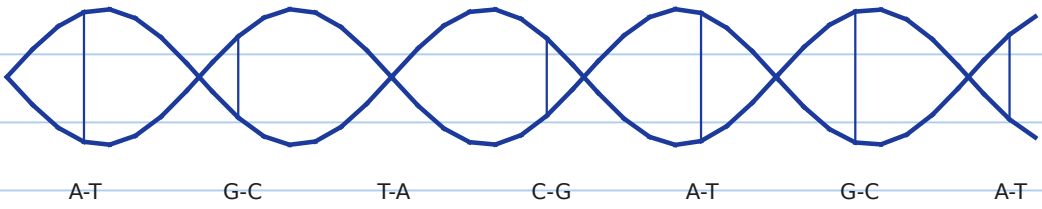
Question: 16. vi) Enlist enzymes use in rDNA technology?

Answer: The enzymes used in recombinant DNA technology include restriction endonucleases, which cut DNA at specific sites, DNA ligase, which joins DNA fragments, and DNA polymerase, which synthesises new DNA strands. Alkaline phosphatase is used to prevent a cut vector from rejoining to itself, and Taq polymerase is used in PCR to amplify DNA.

## 3. Give detailed answers to following questions:

**Question: 17. i) What is DNA sequencing, and how has it been used to study genetic mutations and diseases?**

Answer: DNA sequencing is the technique of determining the exact order of nucleotides (A, T, G and C) in a DNA molecule, most commonly by the Sanger method using chain-terminating nucleotides and gel electrophoresis. To study a disease, the gene of a patient is sequenced and compared with the normal sequence; any change, such as substitution, deletion or insertion of bases, identifies the mutation causing the disorder, as in cystic fibrosis, sickle cell anaemia and many cancers. Sequencing has also allowed carrier detection, prenatal diagnosis, discovery of disease genes through the Human Genome Project and the choice of suitable treatment for individual patients.




Sugar-phosphate backbones with base pairs joined by hydrogen bonds

Fig.: Dna (labelled diagram)

**Question: 18. ii) What is the role of DNA analysis in forensic science and paternity testing, and what are some of the methods used in this field?**

Answer: DNA analysis identifies a person from a very small sample such as blood, hair, semen or skin cells, because the pattern of DNA markers of every person (except identical twins) is unique. In forensic science, DNA fingerprinting of evidence from a crime scene is matched with the DNA of suspects to identify the criminal or to prove an innocent person free, while in paternity testing the markers of a child are compared with those of the mother and the alleged father, since half of the child's markers must come from the father. The main methods used are extraction of DNA, its amplification by PCR, cutting with restriction enzymes to obtain RFLPs, separation by gel electrophoresis, and comparison of STR (short tandem repeat) patterns.

Question: 19. iii) Describe the process of creating genomic maps and explain how this technology has been used to study genes and diseases.

Answer: A genomic map shows the positions of genes and other markers on all the chromosomes of an organism. It is created first as a genetic linkage map, in which the distance between markers is estimated from how often they are separated by crossing over in families, and then as a physical map, in which overlapping DNA fragments are cloned, cut with restriction enzymes and sequenced to fix their exact positions in base pairs. Genomic maps, completed for man in the Human Genome Project, have been used to locate disease genes such as those for cystic fibrosis and Huntington disease, to identify carriers, to compare genomes of different species and to develop new diagnostic tests and gene-based medicines.

# Chapter 27: Biology and Human Welfare

## 1. Encircle the correct answer:

1. i) DT vaccine is used for

(a) Hepatitis

(b) Measles

(c) Common cold

✓ (d) Diphtheria toxoid

Answer: (d) Diphtheria toxoid - DT vaccine contains diphtheria (and tetanus) toxoid, that is the inactivated toxin of the bacteria. It stimulates the body to make antibodies that protect against diphtheria.

2. ii) Clostridium tetani which caused tetanus is a

(a) Virus

✓ (b) Bacteria

(c) Parasite

(d) Fungi

Answer: (b) Bacteria - Clostridium tetani is a rod-shaped bacterium found in soil and dust that enters the body through wounds. Its toxin affects the nervous system and causes the painful muscle stiffness of tetanus.

3. iii) MMR vaccine is effective for

(a) Measles

(b) Polio

✓ (c) Measles, Mumps & rubella

(d) Hepatitis

Answer: (c) Measles, Mumps & rubella - MMR is a combined vaccine whose

name is made from the first letters of the three viral diseases it prevents: measles, mumps and rubella. It is given to children to provide immunity against all three diseases together.

4. iv) The top milk producing animal of Pakistan is

(a) Bhadawari

(b) Jaffrabadi

✓ (c) Nili- Ravi

(d) Godavari

Answer: (c) Nili- Ravi - The Nili-Ravi buffalo, a famous breed of Punjab, is the top milk producing animal of Pakistan. A good animal gives a high daily yield of milk with high fat content, so it is most valued by dairy farmers.

5. v) The yeast that gives puff appearance to dough is called

✓ (a) Saccharomyces

(b) Lactobacillus

(c) Penicillium

(d) Escherichia coli

Answer: (a) Saccharomyces - The yeast Saccharomyces cerevisiae ferments sugars in the dough and releases carbon dioxide gas. The trapped gas bubbles make the dough rise and give it a puffy, spongy appearance, which is used in making bread.

6. vi) The removal of floating solids and organic materials from the sewage is

✓ (a) Primary treatment

(b) Secondary treatment

(c) Tertiary treatment

(d) none of these

Answer: (a) Primary treatment - Primary treatment is the physical first stage of sewage treatment, in which floating and suspended solids are removed by screening, filtration and sedimentation. The settled solid material forms sludge and the liquid passes on for biological treatment.

7. vii) The process of hybrid crossing with its parents is called

(a) Selection

(b) Hybridization

✓ (c) Back cross

(d) Acclimatization

Answer: (c) Back cross - When an F1 hybrid is crossed with either of its parents, the cross is called a back cross. Breeders use it to transfer a desired character of the hybrid into the parent variety while keeping most of the parent's good qualities.

8. viii) French beans and Cauliflowers are vegetables of

✓ (a) Winter

(b) Summer

(c) Spring

(d) all seasons

Answer: (a) Winter - French beans and cauliflower grow best in cool weather, so in Pakistan they are sown and harvested as winter vegetables. High summer heat spoils their growth and, in cauliflower, prevents proper curd formation.

9. ix) The Science that deals with the modification of DNA and enhances

the characteristic of an organism is called

(a) Genetic Engineering

(b) Biotechnology

(c) Biology

(d) None of these

Answer: (a) Genetic Engineering - Genetic engineering directly changes the DNA of an organism by removing, altering or adding genes. In this way a desired characteristic is enhanced or a new useful characteristic is introduced into the organism.

10. x) Sludge digestion is a biological process in which organic solids are decomposed into stable substances

(a) Stable substances

(b) non stable substances

(c) Effluent

(d) None of these

Answer: (a) Stable substances - During sludge digestion, anaerobic microbes break down the organic matter of sludge into stable, harmless substances, along with gases such as methane. The digested sludge can then be dried and safely used as manure.

## 2. Write short answer of the following

Question: 11. i) Why vaccination is important for infectious disease?

Answer: Vaccination introduces a weakened or killed pathogen, or its toxin, into the body, so the immune system makes antibodies and memory cells against it without causing the disease. When the real pathogen later attacks, the body responds quickly and the person does not become ill. Vaccination of a whole population has controlled or eradicated dangerous

infectious diseases such as smallpox and polio.

Question: 12. ii) List the outstanding milk producing breeds of cow.

Answer: Outstanding milk producing breeds of cow include the local breeds Sahiwal and Red Sindhi, which are well adapted to the hot climate of Pakistan and give good yields. Among the exotic high-yielding breeds, Holstein Friesian is the highest milk producer, and Jersey and Brown Swiss are also kept for their rich, high-fat milk. Crosses of local breeds with these exotic breeds combine high yield with adaptation to local conditions.

Question: 13. iii) What do you mean by integrated diseases and its management?

Answer: Integrated disease management means the planned control of the serious and common diseases of a population by using all suitable measures together, instead of depending on one method alone. It includes vaccination, healthy habits and sanitation, clean water and food, early diagnosis and treatment, and biological control using other living organisms. Used together, these measures reduce the spread and severity of diseases effectively and economically.

Question: 14. iv) What is the role of livestock?

Answer: Livestock animals such as cow, buffalo, goat, sheep and poultry provide highly nutritious food in the form of milk, meat and eggs, which supply proteins, fats and vitamins. They also provide wool, leather and other products, draught power for farming and transport, and manure for fields. Livestock is therefore a major source of income and employment for rural families and an important part of the national economy.

Question: 15. v) Differentiate between following (a) Selection and Backcross (b) Vegetables of summer and vegetables of winter (c) Selection and Hybridization

Answer: (a) Selection is the choosing of plants or animals with desired characters for breeding, while a back cross is the crossing of an F1 hybrid with one of its parents. (b) Summer vegetables, such as okra (bhindi), bitter gourd, bottle gourd and brinjal, grow in hot weather and are sown in spring or summer, while winter vegetables, such as cauliflower, cabbage, carrot, radish and peas, grow in cool weather and are sown in autumn or winter. (c) Selection improves a breed by repeatedly breeding from the best existing individuals of the same variety, while hybridization produces a new hybrid by crossing two different varieties or breeds to combine their desired characters.

## 3. Give detail answer of the following questions

Question: 16. i) Explain in detail different steps of sewage treatment.

Answer: Sewage treatment is done in three main steps. In primary treatment, sewage is passed through screens and sedimentation tanks, where floating and suspended solids are removed physically and settle down as sludge, while the clear liquid is called effluent. In secondary treatment, which is a biological process, air is bubbled through the effluent and aerobic microbes decompose the dissolved organic matter, greatly reducing its harmful content; the microbes then settle as activated sludge, part of which is used again and part sent for sludge digestion by anaerobic bacteria, which also produce biogas. In tertiary treatment, the water is further purified by chemical methods and disinfection, for example with chlorine or ultraviolet light, to remove remaining nutrients, pathogens and chemicals before the water is released or reused.

Question: 17. ii) Explain vaccination and its importance.

Answer: Vaccination is the process of introducing a vaccine, made from weakened or killed microbes or their inactivated toxins, into the body to protect it against a specific disease, a method first developed by Edward Jenner in 1796 using cowpox against smallpox. The vaccine acts as an antigen: the body produces antibodies and memory cells, so on later natural infection the immune response is rapid and strong and the disease is prevented. Its importance is very great: vaccination has eradicated smallpox, has nearly eradicated polio, and controls diseases such as measles, diphtheria, tetanus, hepatitis and COVID-19, saving millions of lives, especially of children, and reducing the cost of treatment.

Question: 18. iii) Describes the role of microbes in food processing and sewage treatments.

Answer: In food processing, microbes are used to make many everyday foods: Lactobacillus converts milk into curd and yoghurt, the yeast Saccharomyces cerevisiae raises bread dough and produces alcohol, and other microbes are used in cheese, vinegar and fermented foods, improving their taste, texture and keeping quality. In sewage treatment, microbes do the main cleaning work: aerobic bacteria and other microorganisms in secondary treatment decompose the organic pollutants of sewage into harmless substances, and anaerobic bacteria digest the sludge, producing methane gas that can be used as fuel. In both fields microbes convert waste or raw material into useful, safe products.